



Chapter 2

The First Law



Thermodynamics

- The principle of energy changes
- The first law: the transformation of energy
- The second law: the origin of the spontaneity of physical and chemical change



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The basic concepts

Matter

- **Substance:**

A substance is a distinct, pure form of matter.

- **Intensive property 强度性质:**

A property that is independent of the amount of the substance.

- **Extensive property 广度性质:**

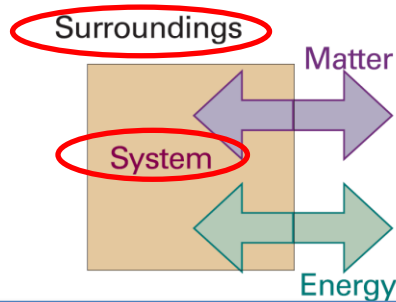
A property that depends on the amount of the substance.



Systems and surroundings

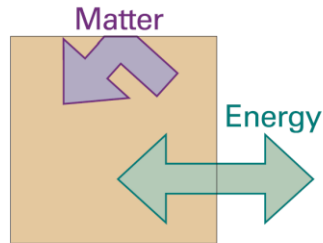
- For the purposes of physical chemistry, the universe is divided into two parts, the **system** and its **surroundings**.
- The **system** is the part of the world in which we have a special interest.
- The **surroundings** are where we make our measurements.
- The type of system depends on the characteristics of the boundary that divides it from the surroundings.

2.1 Work, heat, and energy



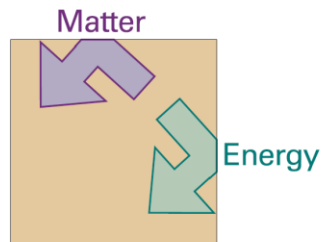
(a) Open

(a). Open system
can exchange matter and energy with its surroundings.



(b) Closed

(b). Closed system
can exchange energy with its surroundings, but it cannot exchange matter.



(c) Isolated

(c). Isolated system
can exchange neither energy nor matter with its surroundings.



Energy

The central concept of all explanations in physical chemistry is that of energy. Briefly, **energy is the capacity to do work.**

There are two contributions to the total energy of a system from the matter it contains. The **kinetic energy** and the **potential energy**.



Energy

1). Kinetic energy:

the energy it possesses as a result of its motion.

2). Potential energy:

the energy it possesses as a result of its position.

3). The unit of energy:

Joule (J), SI unit; $1 \text{ J} = 1 \text{ N} \cdot \text{m} = \text{Pa} \cdot \text{m}^3 = 1 \text{ kg m}^2 \text{ s}^{-2}$.

① Energy transfer: heat, work

② Depended by processes, not state function

- Occurring when the state of the system changes
- Different processes have different values of work and/or heat

Heat

When the energy of a system changes as a result of temperature difference between the system and its surroundings, q

Work

Work is done when an object is moved against an opposing force.

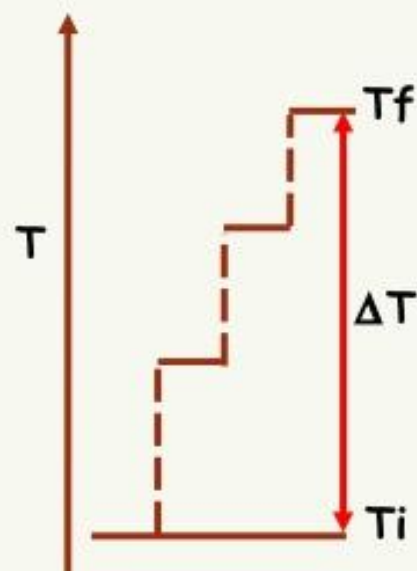
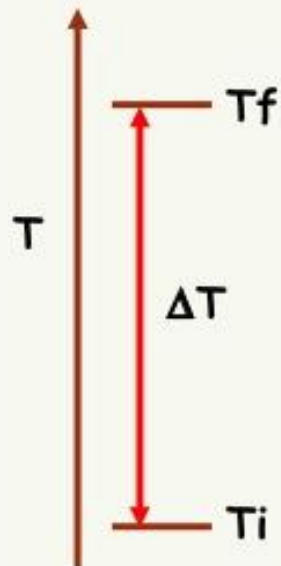
Energy other than heat, w

Unit of q & w : J.

Not state function, the value relates to the path

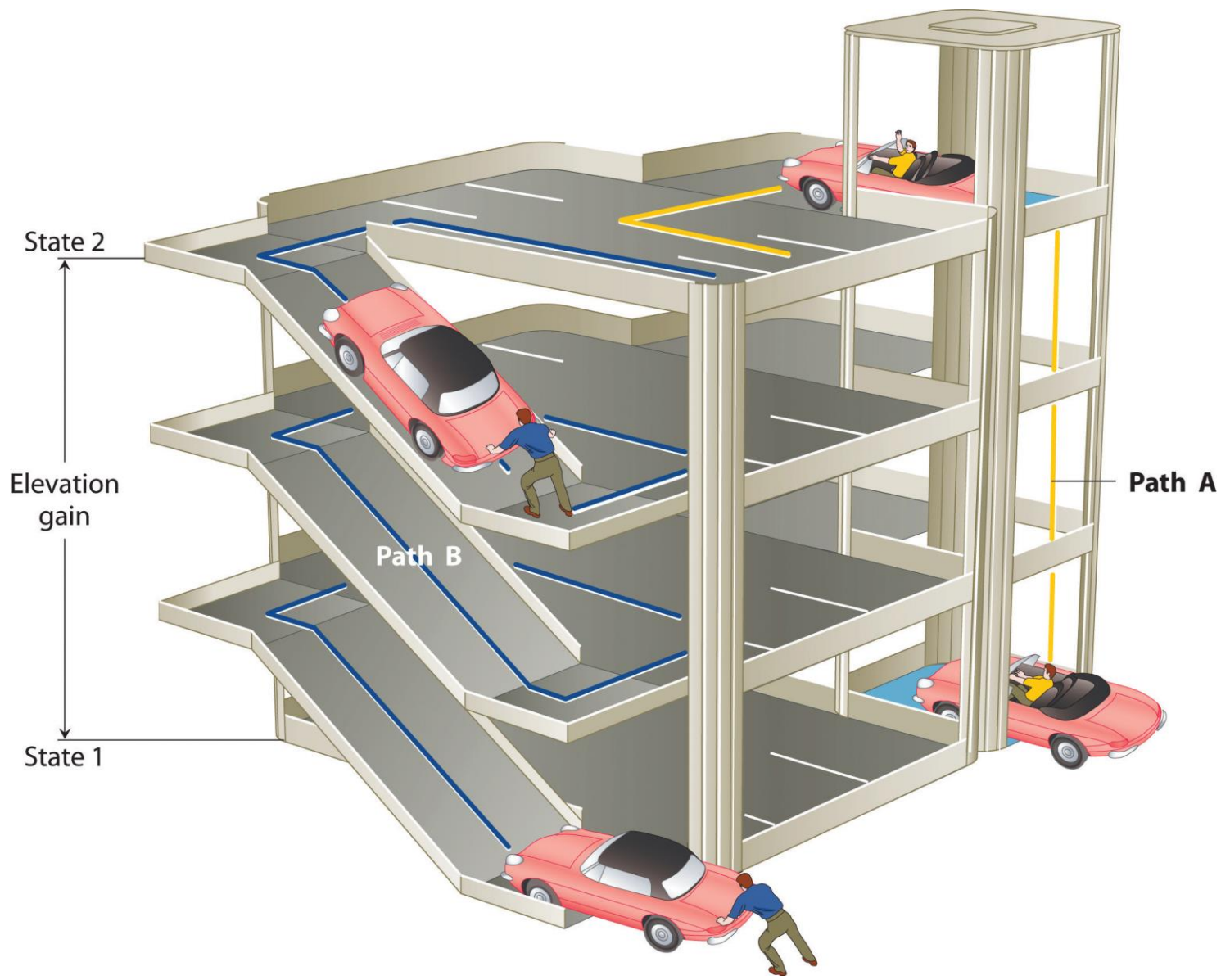


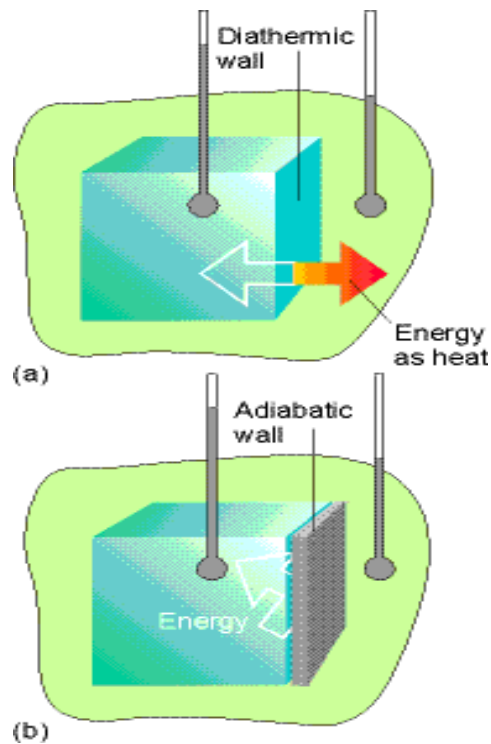
A Thermodynamic "State Function"



A State Function is defined as a property of a system that only depends on the initial and final state of the system,

and is independent of the path followed in getting from one to the other.



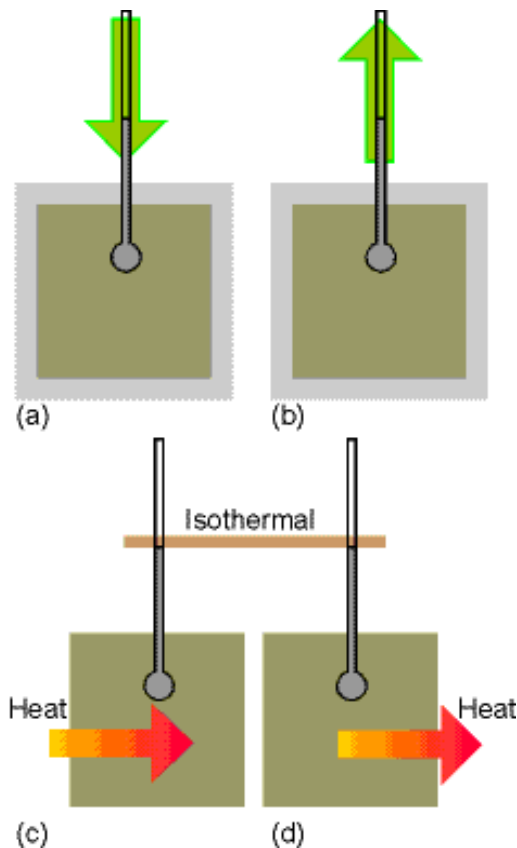


(a). A **diathermic system** is one that allows energy to escape as heat through its boundary if there is a difference in temperature between the system and its surroundings.

(b). An **adiabatic system** is one that does not permit the passage of energy as heat through its boundary even if there is a temperature difference between the system and its surroundings.



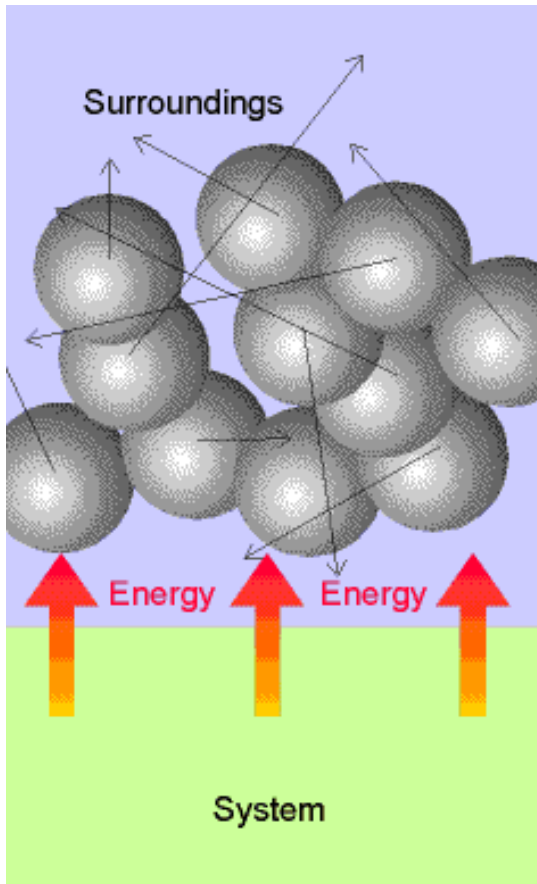
Exothermic process: one that releases energy as heat.
Endothermic process: one that absorbs energy as heat.



(a) When an endothermic process occurs in an adiabatic system, the temperature falls; (b) if the process is exothermic, then the temperature rises. (c) When an endothermic process occurs in a diathermic container, energy enters as heat from the surroundings, and the system remains at the same temperature; (d) if the process is exothermic, then energy leaves as heat, and the process is isothermal.



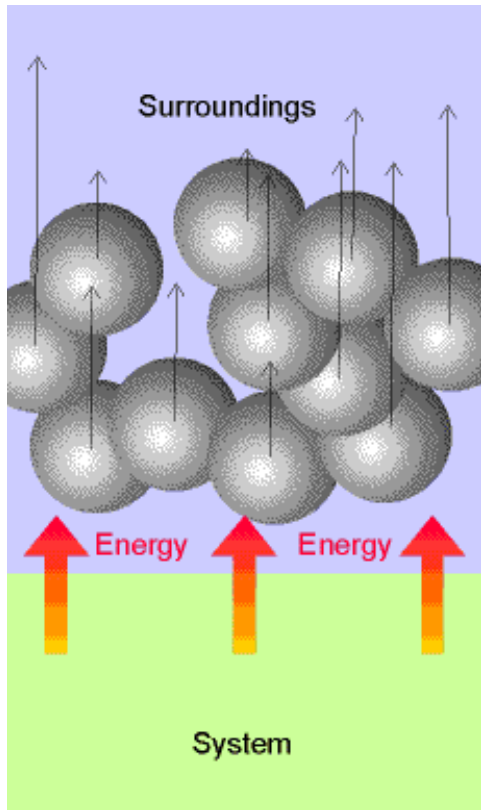
Heat-molecular interpretation



Heat is the transfer of energy that makes use of chaotic molecular motion. The chaotic motion of molecules is called **thermal motion**. When a system heats its surroundings, molecules of the system stimulate the thermal motion of the molecules in the surroundings.



Work-molecular interpretation



Work is the transfer of energy that makes use of organized motion. When a system does work, it stimulates orderly motion in the surroundings



2.2 The first law

The internal energy, U

- The total energy of a system is called its internal energy;
- It is the total kinetic and potential energy of the molecules composing the system.
- The internal energy is a state function. It is an extensive property.
- Internal energy, heat, and work are measured in the same units, the Joule (J).



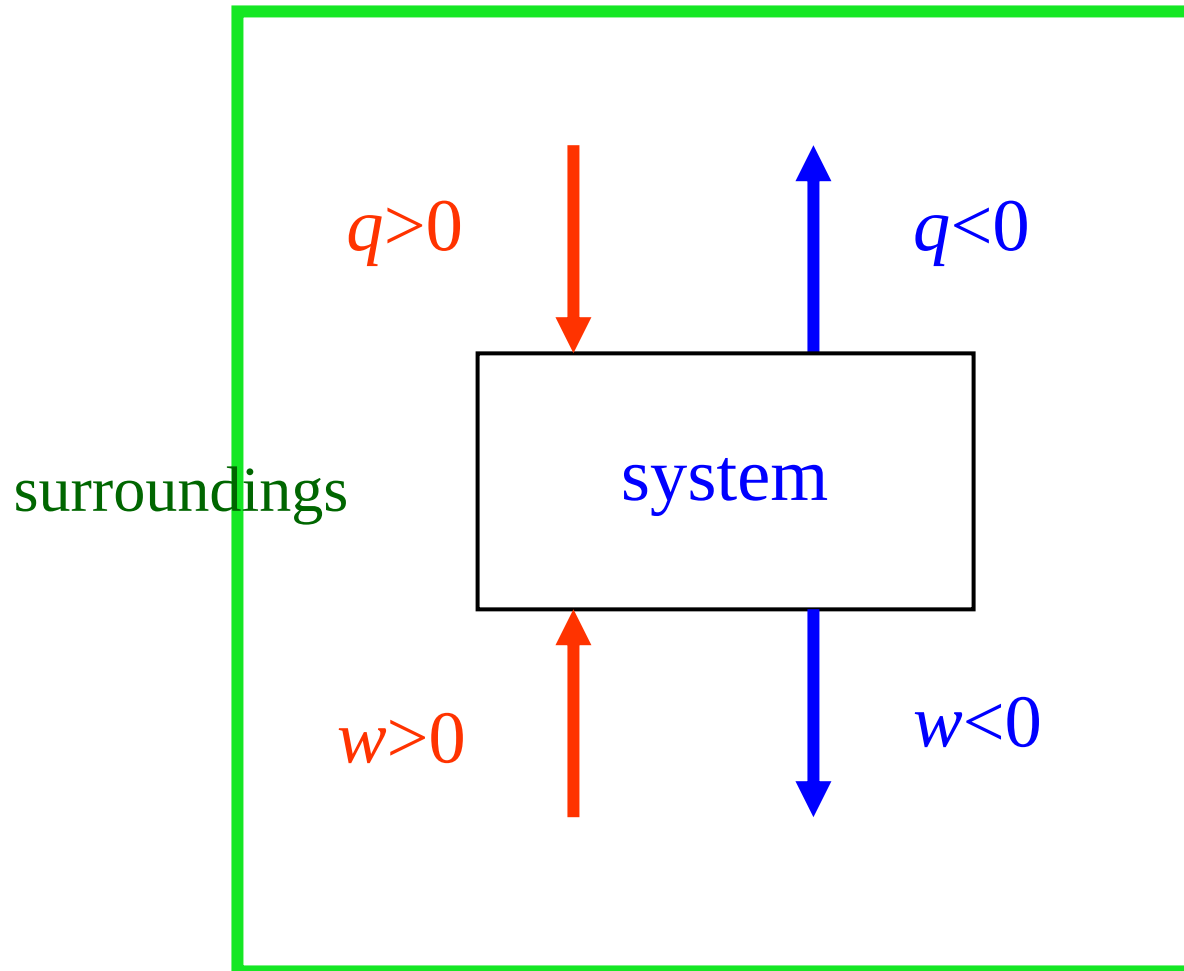
The conservation of energy

- Heat and work are equivalent ways of changing a system's internal energy.
- If we write w for the work done on a system, q for the energy transferred as heat to a system, and ΔU for the resulting change in internal energy, then it follows that

The first law of thermodynamics

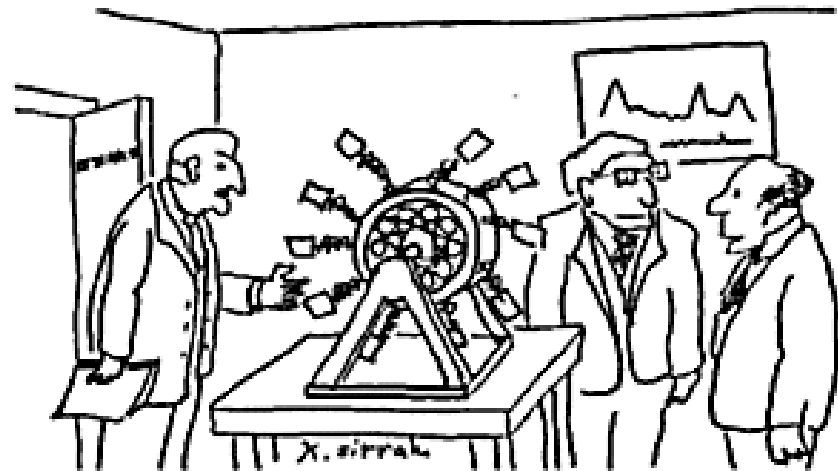
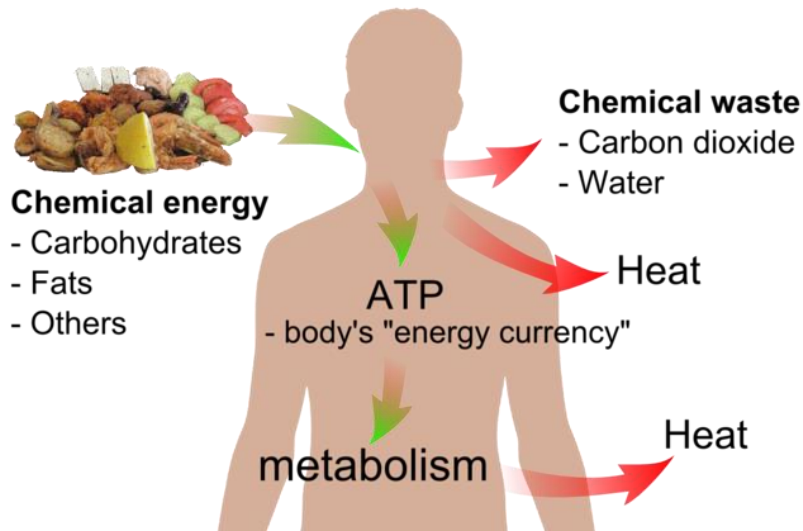
The internal energy of an **isolated system** is constant

$$\Delta U = q + w$$



$w > 0$ or $q > 0$, energy is transferred to the system as work or heat.
 $w < 0$ or $q < 0$, energy is lost from the system as work or heat.

Energy and human life



"It may be perpetual motion, but it will take forever to test it"
Cartoon by Donald Simanek

We cannot use a system to do work, leave it isolated for a month, and then come back expecting to find it restored to its original state and ready to do the same work again. The evidence for this property is that no 'perpetual motion machine' (a machine that does work without consuming fuel or some other source of energy) has ever been built.



2.3 Expansion work

Expansion work: The work arising from a change in volume

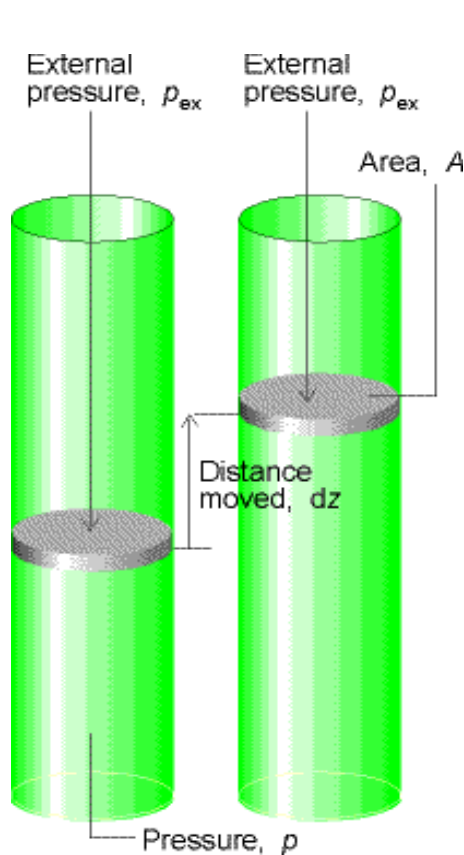
A. The general expression for work

General work is defined as the work which required to move an object a distance dz against an opposing force of magnitude F is

$$dw = -Fdz$$

The **negative sign** presents that when the system moves an object against an opposing force, the internal energy of the system doing the work will decrease.

The wall of the system is a massless, frictionless, rigid, perfectly fitting piston of area A



the area of the piston, A

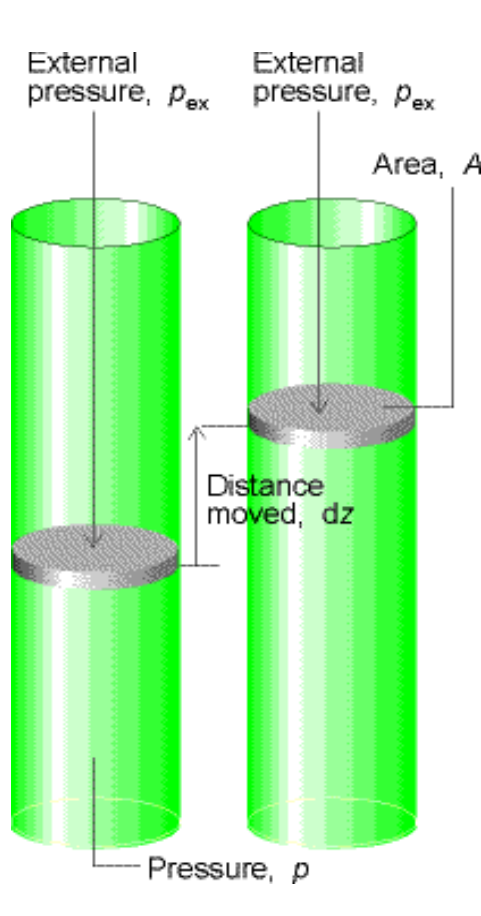
the external pressure, p_{ex}

the force on the outer face, $F = p_{ex}A$.

When the system expands through a distance dz against p_{ex} , the work done:

$$dw = -p_{ex}Adz$$

where Adz is the change in volume, dV , in the course of the expansion.



The work done when the system expands by dV against a pressure p_{ex} is

$$dw = -F dz = -p_{\text{ex}} dV$$

The total work done, w :

$$w = -\int_{V_i}^{V_f} p_{\text{ex}} dV$$



Other types of work (for example, electrical work), which we shall call either **non-expansion work** or **additional work**, have analogous expressions, with each one the product of an intensive factor (the pressure, for instance) and an extensive factor (the change in volume).

Table 2.1 Varieties of work*

Type of work	dw	Comments	Units†
Expansion	$-p_{\text{ex}}dV$	p_{ex} is the external pressure dV is the change in volume	Pa m^3
Surface expansion	$\gamma d\sigma$	γ is the surface tension $d\sigma$ is the change in area	N m^{-1} m^2
Extension	$f dl$	f is the tension dl is the change in length	N m
Electrical	ϕdQ	ϕ is the electric potential dQ is the change in charge	V C

* In general, the work done on a system can be expressed in the form $dw = -Fdz$, where F is a 'generalized force' and dz is a 'generalized displacement'.

† For work in joules (J). Note that $1 \text{ N m} = 1 \text{ J}$ and $1 \text{ V C} = 1 \text{ J}$.



B. Free expansion

Free expansion: the expansion against zero opposing force.

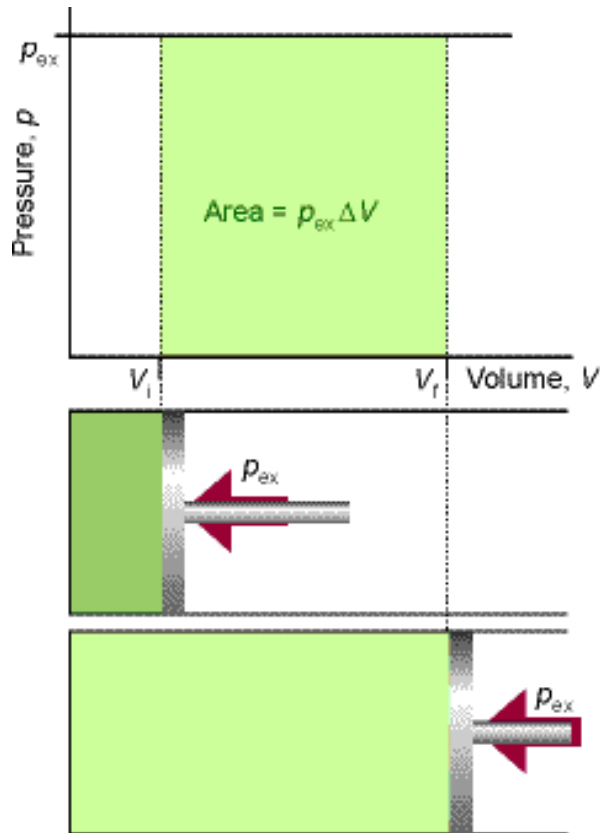
$$p_{\text{ex}} = 0 \quad \rightarrow \quad dw = 0 \text{ for each stage of the expansion}$$

$$w = 0$$

- **No work** is done when a system expands freely.
- Expansion of this kind occurs when a system **expands into a vacuum.**

C. Expansion against constant pressure

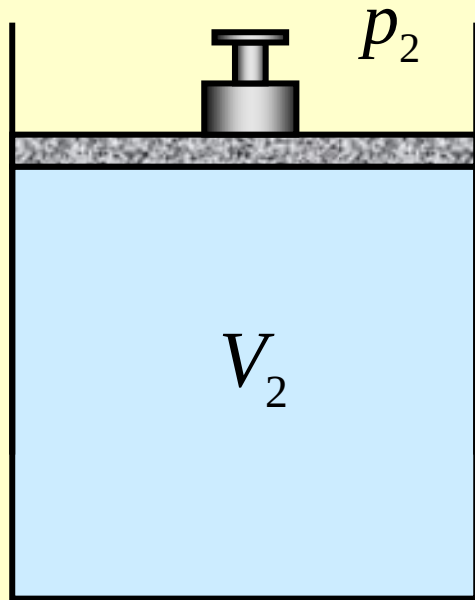
For an expansion of the external pressure constant throughout of the expansion



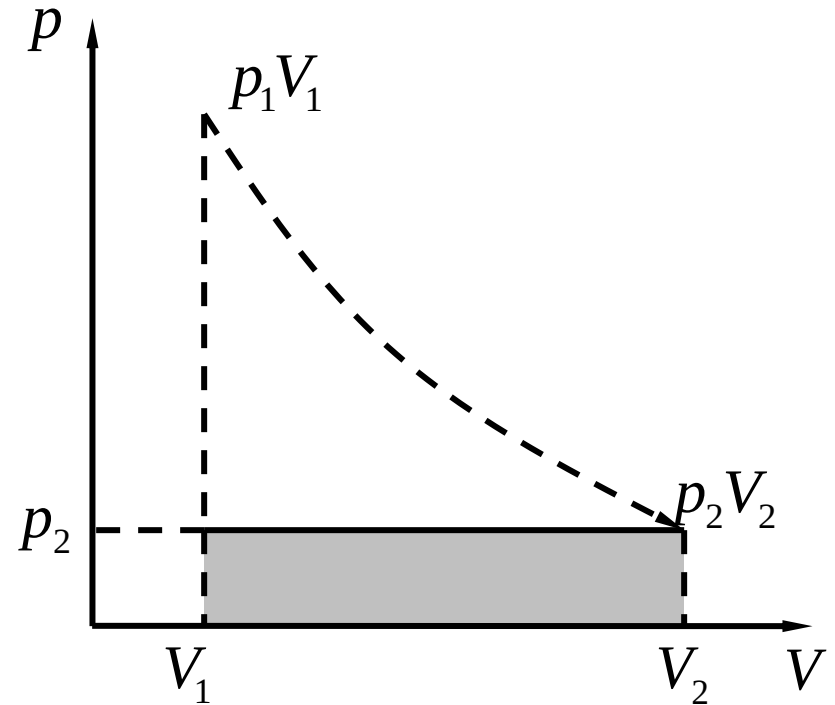
$$w = -p_{\text{ex}} \int_{V_i}^{V_f} dV$$

$$= -p_{\text{ex}} (V_f - V_i) = -p_{\text{ex}} \Delta V$$

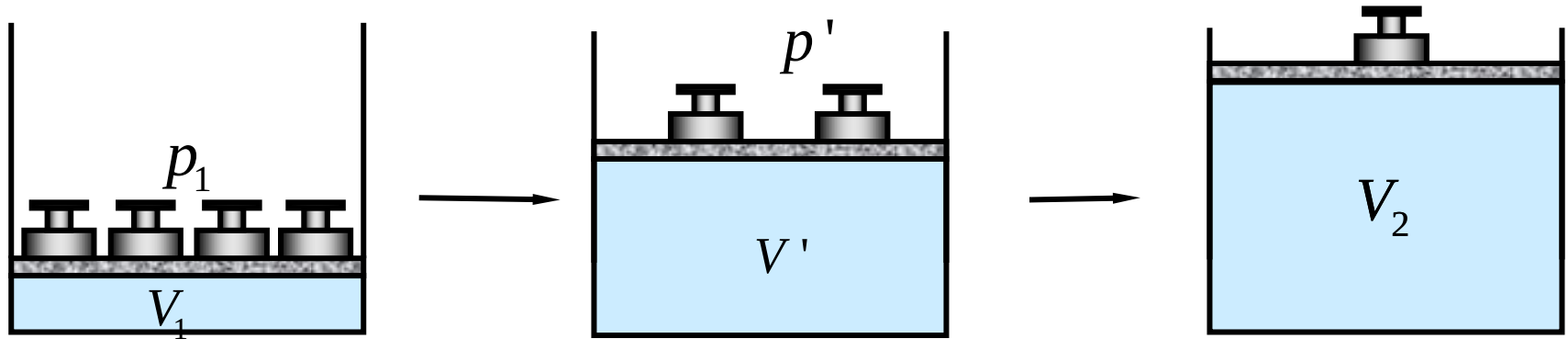
The magnitude of w , is equal to the area beneath the horizontal line at $p = p_{\text{ex}}$ lying between the initial and final volumes.



The shadow area w_e



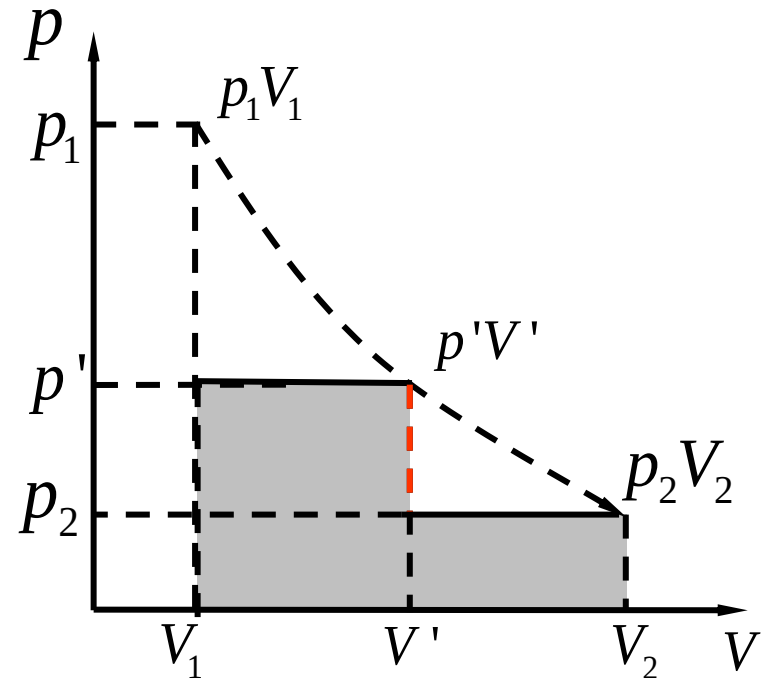
$$w_e = -p_2(V_f - V_i) = -p_2\Delta V$$

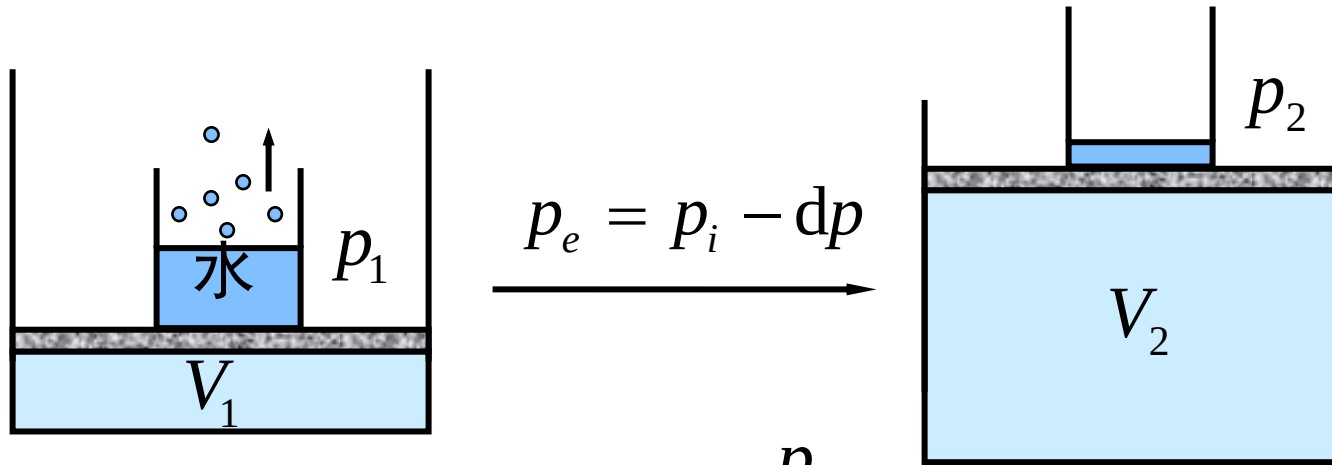


(1) p' volume from V_1 to V'

(2) p_2 volume from V' to V_2

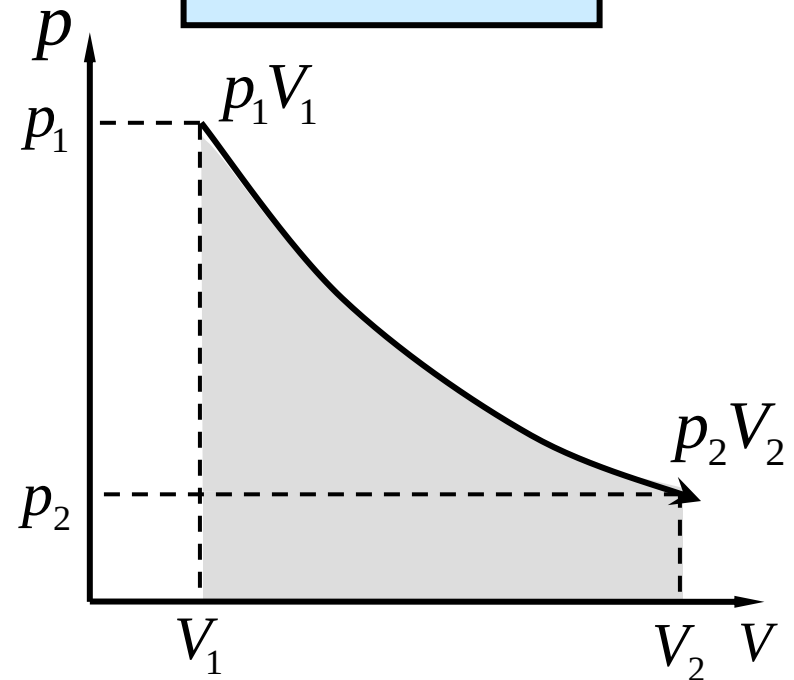
$$w_e = -p'(V' - V_1) - p_2(V_2 - V')$$

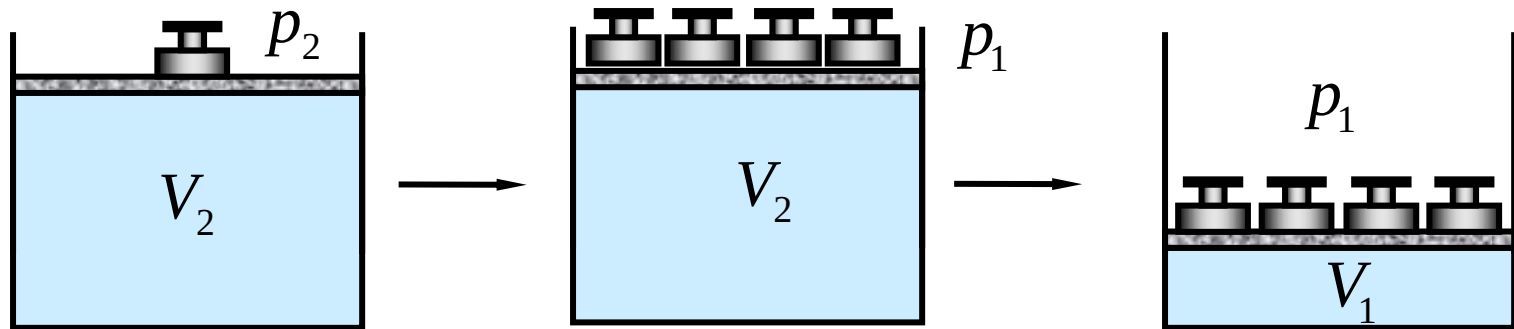




$$dw = -p_e dV = -pdV$$

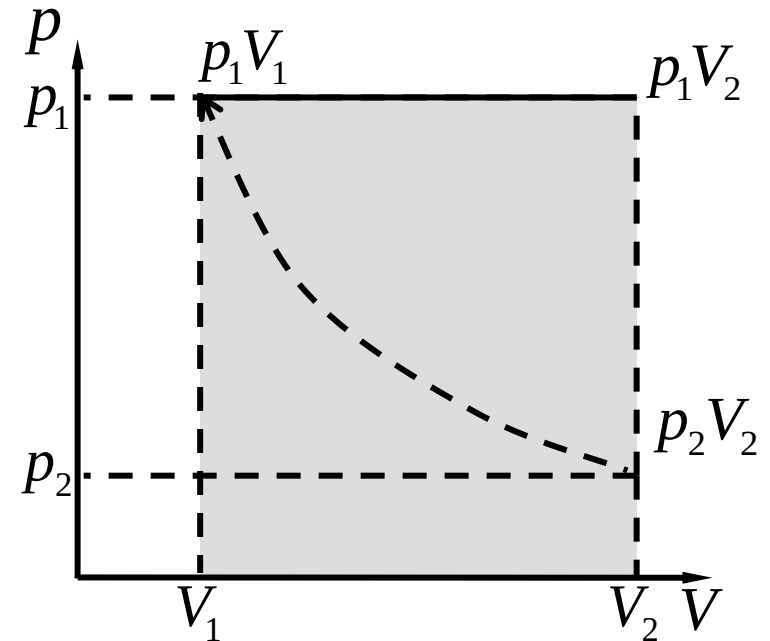
$$W = -\int_{V_i}^{V_f} pdV$$

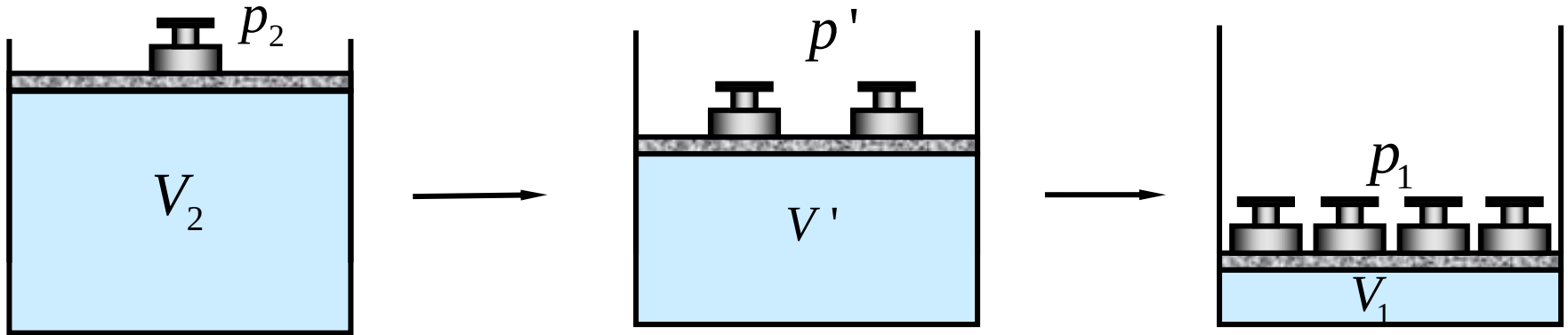




From V_2 to V_1

$$w_e = -p_1(V_1 - V_2)$$

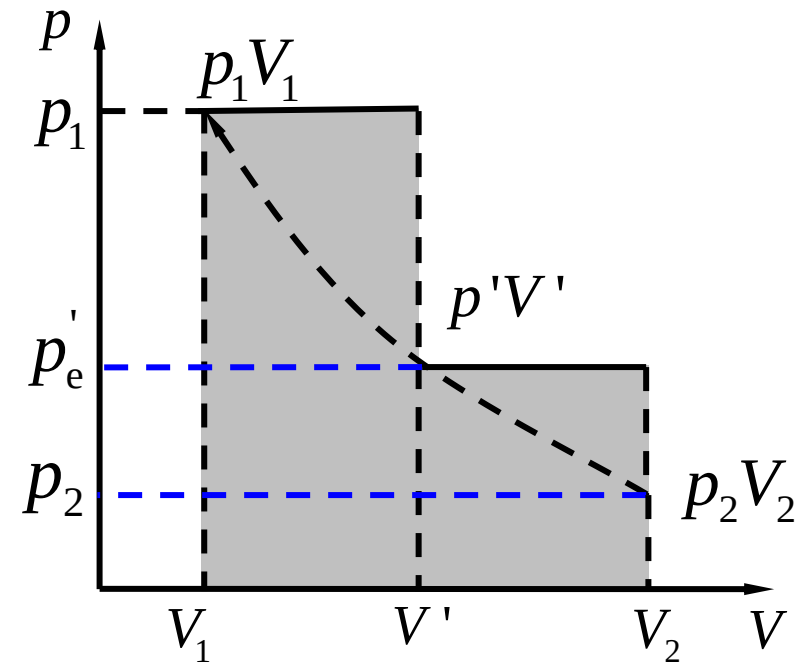


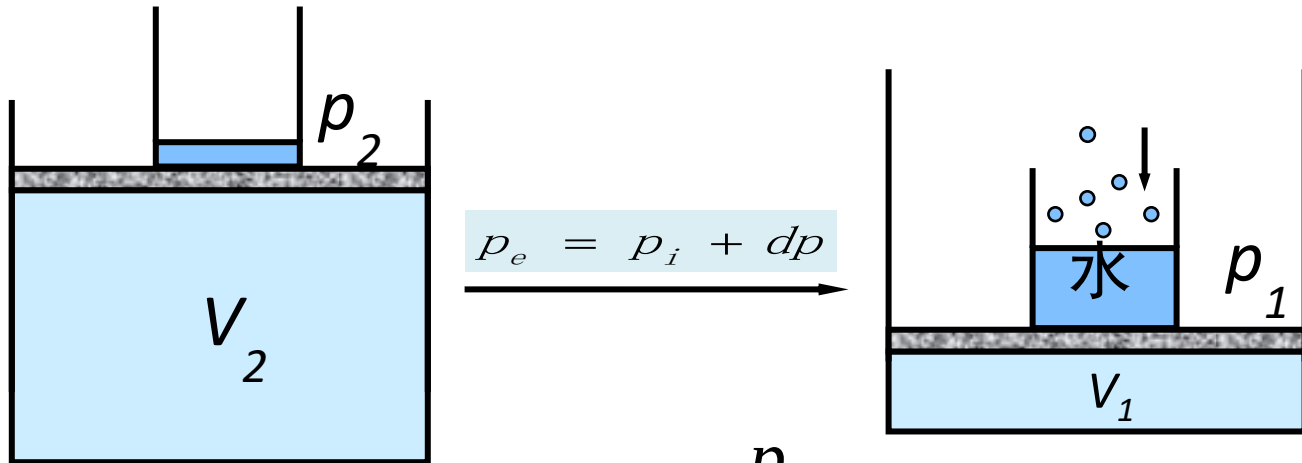


(1) p' volume from V_2 to V'

(2) p_1 volume from V' to V_1

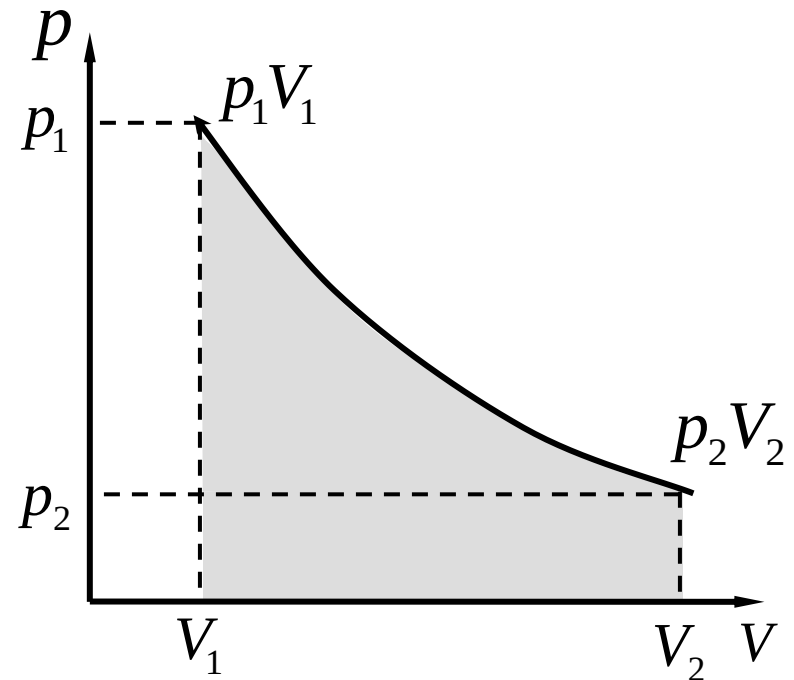
$$w_e = -p'(V' - V_2) - p_1(V_1 - V')$$

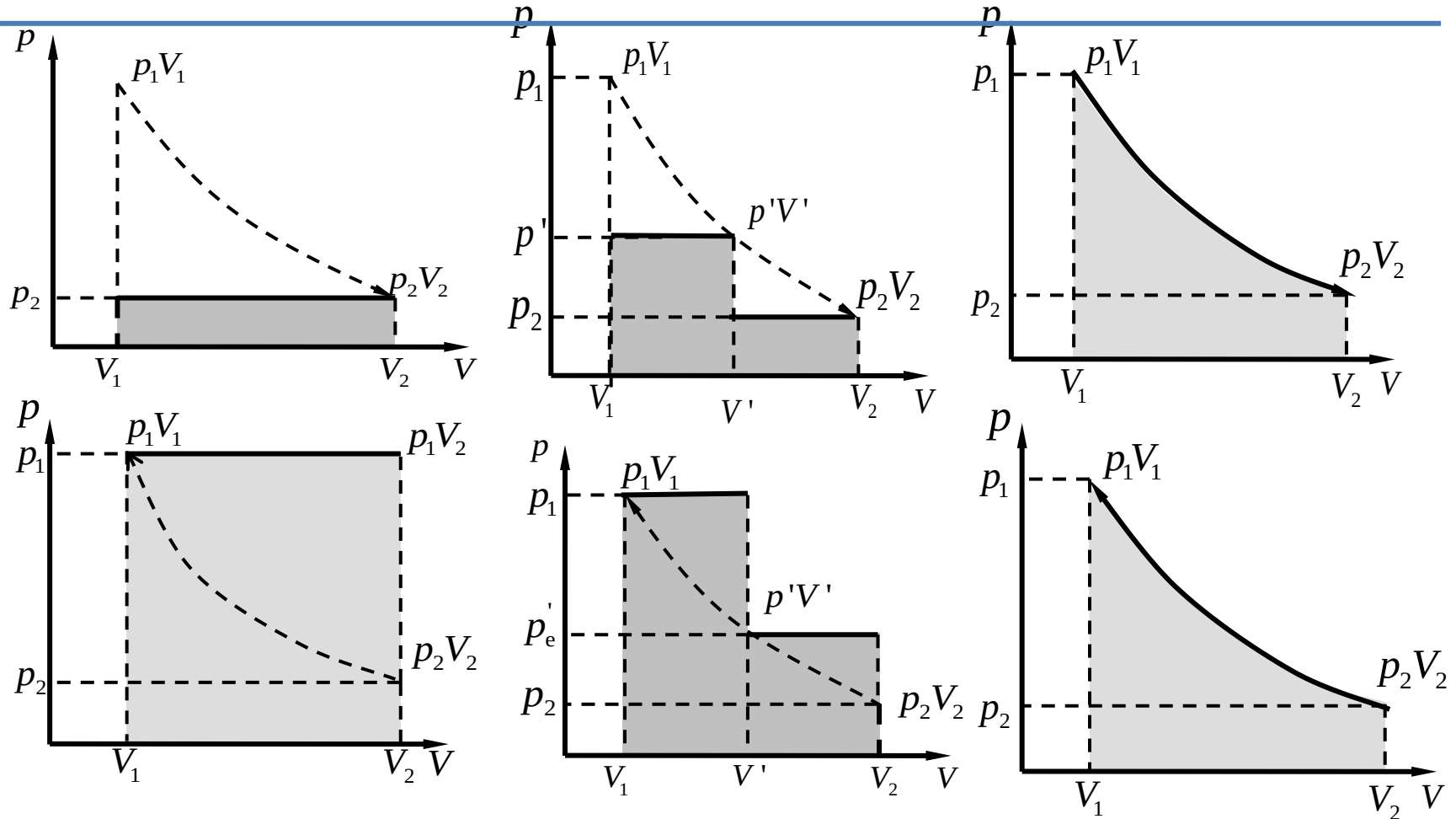




$$dw = -p_{\text{ex}} dV = -pdV$$

$$W = -\int_{V_i}^{V_f} pdV$$





Reversible expansion, system does the maximal work to surrounding.

Reversible compression, surrounding does the minimal work to system.

Reversible expansion follows reversible compression, no changes left in system and surrounding.



D. Reversible expansion

A **reversible** change in thermodynamics is a change that can be reversed by an infinitesimal modification of a variable. The key word '**infinitesimal**' sharpens the every-day meaning of the word '**reversible**' as something that can change direction.



To achieve a **reversible expansion**, set the external pressure p_{ex} equal to the pressure of the confined gas p at each stage of the expansion, $p_{\text{ex}} = p - dp$

$$dw_{\text{rev}} = -p_{\text{ex}}dV = -(p - dp)dV = -pdV$$

$$w = -\int_{V_i}^{V_f} p dV$$

This equation is for the reversible expansion. The integral can be evaluated once we know how the pressure of the confined gas depends on its volume.



E. Isothermal reversible expansion

For the isothermal, reversible expansion of a perfect gas:

$$w_{rev,T} = -nRT \int_{V_i}^{V_f} \frac{dV}{V} = -nRT \ln \frac{V_f}{V_i}$$

If in an expansion,

$$V_f > V_i \quad \Rightarrow \quad w_{rev,r} < 0$$

- The system has done work on the surroundings.
- The internal energy of the system has decreased as a result of the work it has done.



Example

Calculate the work done when 50 g of iron reacts with hydrochloric acid in (a) a closed vessel of fixed volume, (b) an open beaker at 25 °C.

Method: We need to judge the magnitude of the volume change, and then to decide how the process occurs. If there is no change in volume, there is no expansion work however the process takes place. If the system expands against a constant external pressure, the work can be calculated. A general feature of processes in which a condensed phase changes into a gas is that the volume of the former may usually be neglected relative to that of the gas it forms.



Answer: In (a) the volume cannot change, so no work is done and $w = 0$. In (b) the gas drives back the atmosphere and therefore $w = -p_{\text{ex}} \Delta V$. If neglect the initial volume, $\Delta V = V_f - V_i \approx V_f = nRT / p_{\text{ex}}$, where n is the amount of H_2 produced.

$$w = -p_{\text{ex}} \Delta V \approx -p_{\text{ex}} \times \frac{nRT}{p_{\text{ex}}} = -nRT$$



The reaction is



Because 1 mol H_2 is generated when 1 mol Fe is consumed

$$w = -nRT$$

$$= -\frac{50\text{g}}{55.85\text{g mol}^{-1}} (8.314 \text{ J K}^{-1} \text{ mol}^{-1})(298.15\text{K})$$

$$= -2.2\text{kJ}$$



Discussion

Try to justify that the one-time expansion is not reversible expansion while the infinitesimal expansion is reversible from the example as following.

$1\text{mol H}_2 (3000\text{Pa}, 1\text{m}^3) \longrightarrow 1\text{mol H}_2 (1000\text{Pa}, 3\text{m}^3)$



1mol H₂ (3000Pa, 1m³) $\xrightarrow[\text{at constant pressure}]{\text{Expansion}}$ 1mol H₂ (1000Pa, 3m³)

1. If the expansion is free expansion

$$w=0$$

2. If at $p_e=1000\text{Pa}$, one time expansion

$$\begin{aligned} w &= -p_e(V_2 - V_1) \\ &= -1000 \times (3 - 1) \text{ J} \\ &= -2000 \text{ J} \end{aligned}$$

3. If isothermal reversible expansion

$$\begin{aligned} w &= -nRT \ln(V_2/V_1) \\ &= -3000 \times 1 \times \ln 3 \text{ J} \\ &= -3296 \text{ J} \end{aligned}$$



Reversible process

- + Reversible, not changing system and surroundings
- + Equilibrium
- + Maximum w from system to surroundings
- + Minimum w from surroundings to system
- + Reversible processes: reversible expansion, reversible heat transfer, reversible phase change, reversible chemical reaction, etc.



2.4 Heat transactions

The change in internal energy of a system generally is

$$dU = dq + dw_{\text{exp}} + dw_e$$

w_{exp} : expansion work

w_e : work in addition to the expansion work. For instance, it might be the electrical work of driving a current through a circuit.



A system kept at constant volume

$$w_{exp} = 0$$

Also incapable of doing any other kind of work

$$w_e = 0$$

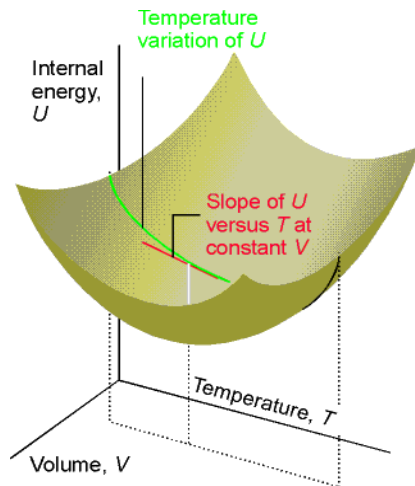
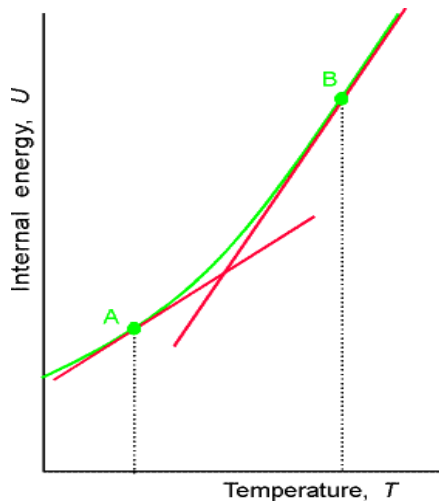
$$\Delta U = q_V$$

$q > 0$: the heat is supplied to a constant-volume system.

$q < 0$: the heat is lost from a constant-volume system.



Heat capacity



The internal energy of a substance increases when its temperature is raised. The slope of the curve at any T is called the **heat capacity, C** , of the system at that T .

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V$$

系统升高单位热力学温度时所吸收的热



C_V – Extensive properties; $J \cdot K^{-1}$

$C_{V,m}$ – Intensive properties; $J \cdot K^{-1} \cdot mol^{-1}$

Heat capacities are extensive properties: 100 g of water, for instance, has 100 times the heat capacity of 1 g of water (and therefore requires 100 times the energy as heat to bring about the same rise in temperature).

The **molar heat capacity at constant volume**, $C_{V,m} = C_V/n$, is the heat capacity per mole of material, and is an intensive property (all molar quantities are intensive).



A change in internal energy to a change in temperature of a constant-volume system follows:

$$dU = C_V dT$$

An infinitesimal change in T brings about an infinitesimal change in U , and the constant of proportionality is C_V at constant V . If C_V is independent of T over the range of temperatures of interest, a measurable change in internal ΔU

$$\Delta U = C_V \Delta T$$



Because a change in U can be identified with the heat supplied at constant V , the above equation is then

$$q_V = C_V \Delta T$$

This relation provides a simple way of measuring C_V of a sample. The ratio of the heat supplied to T rise it causes is the **heat capacity** of the sample.



A given q is transferred to a system.

If C_V is small, ΔT is large

If C_V is large, ΔT is small

If C_V is infinite large, there will be no increase in temperature.

At a phase transition, such as at the boiling point of water, the temperature of a substance does not rise as energy is supplied as heat: the energy is used to drive the endothermic transition, in this case to vaporize the water, rather than to increase its temperature. Therefore, at the temperature of a phase transition, the heat capacity of a sample is infinite.

2.5 Enthalpy

The change in internal energy is not equal to the heat supplied when the system is free to change its volume. Under these circumstances some of the energy supplied as heat to the system is returned to the surroundings as expansion work. However, we shall now show that in this case the heat supplied at constant pressure is equal to the change in another thermodynamic property of the system, the **enthalpy**, H .

A. The definition of enthalpy

As the pressure is constant

- q_p is transferred to the system at constant p
- The system expanded from V_1 to V_2 against constant p

$$\Delta U = U_2 - U_1 = q_p + w = q_p - p(V_2 - V_1)$$

$$q_p = (U_2 + pV_2) - (U_1 + pV_1)$$

A. The definition of enthalpy

The enthalpy, a state function, is defined as:

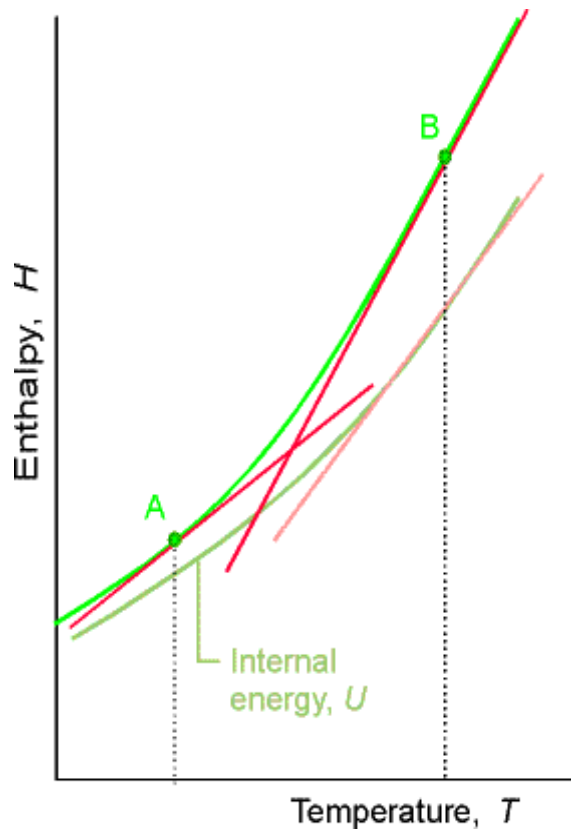
$$H = U + pV$$

- The change in enthalpy between any pair of initial and final states is independent of the path between them.
- ΔH is equal to the heat supplied at constant pressure to a system.

$$q_p = (U_2 + pV_2) - (U_1 + pV_1) = \Delta H$$

$$\Delta H = q_p \text{ (at constant pressure, no additional work)}$$

B. The variation of enthalpy with temperature



The H of a substance increases as its T is raised. The relation between the increase in H and the increase in T depends on the conditions. The most important condition is constant p , and the slope of a graph of H against T at constant p is called the heat capacity at constant p , C_p

$$C_p = \left(\partial H / \partial T \right)_p$$



For infinitesimal changes of temperature:

$$dH = C_p dT$$

If the heat capacity is constant over the range of temperatures of interest:

$$\Delta H = C_p \Delta T$$

$$q_p = C_p \Delta T$$

C_p – Extensive properties; $J \cdot K^{-1}$

Molar heat capacity at constant pressure:

$C_{p,m}$ – Intensive properties; $J \cdot K^{-1} \cdot mol^{-1}$



$$q_p = C_p \Delta T$$

- The variation of heat capacity with temperature can sometimes be ignored if the temperature range is small; this approximation is highly accurate for a monatomic perfect gas (for instance, one of the noble gases at low pressure).
- However, when it is necessary to take the variation into account, a convenient approximate empirical expression is

$$C_{p,m} = a + bT + \frac{c}{T^2}$$

Synoptic Table 2.2* Temperature variation of molar heat capacities, $C_{p,m}/(\text{J K}^{-1} \text{ mol}^{-1}) = a + bT + c/T^2$

	a	$b/(10^{-3} \text{ K})$	$c/(10^5 \text{ K}^2)$
C(s, graphite)	16.86	4.77	-8.54
CO ₂ (g)	44.22	8.79	-8.62
H ₂ O(l)	75.29	0	0
N ₂ (g)	28.58	3.77	-0.50

* More values are given in the *Data section*.

Example - Relating ΔH and ΔU

The internal energy change when 1.0 mol CaCO_3 in the form of calcite converts to aragonite is +0.21 kJ. Calculate the difference between the enthalpy change and the change in internal energy when the pressure is 1.0 bar given that the densities of the solids are 2.71 gcm^{-3} and 2.93 gcm^{-3} , respectively.

Calcium Carbonate (CaCO_3) has two crystalline solid state forms,
 $\text{CaCO}_3(\text{calcite}) \rightarrow \text{CaCO}_3(\text{aragonite})$
 ΔU for the conversion of one mole of calcite to aragonite is +0.21 kJ
Calculate ΔH for this conversion at 1 bar ($1 \times 10^5 \text{ Pa}$).

Example - Relating ΔH and ΔU

Answer: The change in enthalpy when the transformation occurs is

$$\Delta H = H(\text{aragonite}) - H(\text{calcite})$$

$$= \{U(a) + pV(a)\} - \{U(c) + pV(c)\}$$

$$= \Delta U + p\{V(a) - V(c)\}$$

$$= \Delta U + p\Delta V$$

Example - Relating ΔH and ΔU

For the volume of 1.0 mol CaCO_3 (100 g) and $\rho = M/V_m$:

$$V(a) = 100(\text{g}) / 2.93(\text{gcm}^{-3}) = 34\text{cm}^3$$

$$V(c) = 100(\text{g}) / 2.71(\text{gcm}^{-3}) = 37\text{cm}^3$$

$$\begin{aligned} p\Delta V &= 1.0 \times 10^{-5} \text{ Pa} \times (34 - 37) \times 10^{-6} \text{ m}^3 \\ &= -0.3 \text{ Pa m}^3 = -0.3 \text{ J} \end{aligned}$$



$$\Delta H - \Delta U = -0.3 \text{ J}$$



Review

Internal energy (U)

$$\Delta U = w + q \quad (1)$$

Enthalpy (H)

$$H = U + PV \quad (2)$$

The equation (2) is always established, no matter whether the pressure is constant or not. But only if the pressure is constant, the following equation (3) can be used.

$$\Delta H = q_p \quad (3)$$

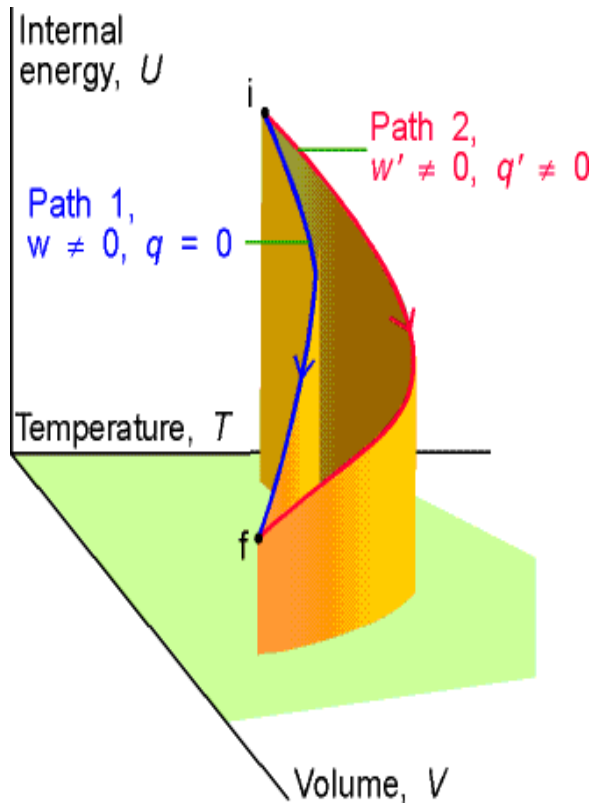
Heat capacity (C)

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V$$

$$C_p = \left(\frac{\partial H}{\partial T} \right)_p$$



State function: Exact and inexact differential

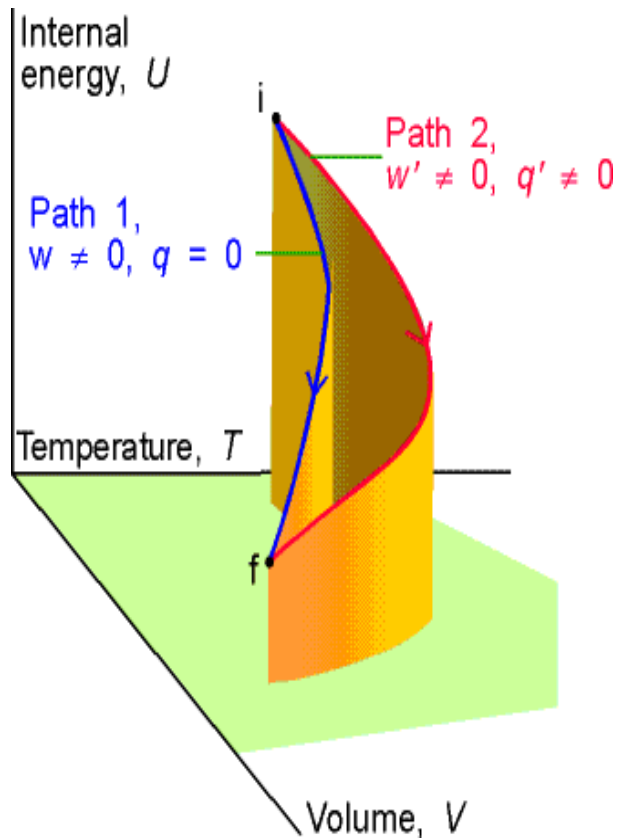


The initial state of the system is **i** and in this state the internal energy is U_i . Work is done by the system as it expands adiabatically to a state **f**. In this state the system has an internal energy U_f and the work done on the system as it changes along **Path 1** from **i** to **f** is w . U is a property of the state; w is a property of the path.

$$\Delta U = \int_i^f dU$$



State function: Exact and inexact differential



In Path 2, the initial and final states are the same but in which the expansion is not adiabatic. In this path an energy q' enters the system as heat and the work w' is not the same as w . The work and the heat are path functions.

$$\Delta U = \int_i^f dU$$



ΔU , independent of the path, an exact differential. An exact differential is an infinitesimal quantity which, when integrated, gives a result that is independent of the path between the initial and final states.

q or w , dependent on the path, an inexact differential. When a system is heated, the total energy transferred as heat is the sum of all individual contributions at each point of the path.

$$\cancel{\Delta q} \quad \cancel{q_f - q_i}$$

$$\underline{q \quad dq}$$



- Work (w)

Free expansion

$$w = 0$$

At a constant pressure

$$w = -p_{ex}\Delta V$$

Reversible

$$w = -\int_{V_i}^{V_f} p dV$$

Isothermal reversible

$$w = -nRT \ln \frac{V_f}{V_i}$$

- Heat (q)

$$q_V = \Delta U$$

$$q_p = \Delta H$$

C. Changes in internal energy

For a closed system of constant composition, U is a function of V and T .

$$U = f(V, T)$$

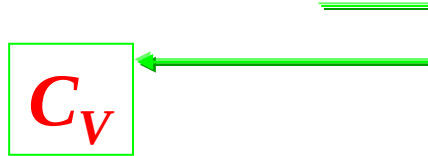
General expression for change in U with V and T is:

$$dU = \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT$$

The interpretation of this equation is that, in a closed system of constant composition, any infinitesimal change in the internal energy is proportional to the infinitesimal changes of volume and temperature, the coefficients of proportionality being the two partial derivatives.



$$dU = \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT$$


$$C_V$$

$\left(\frac{\partial U}{\partial T}\right)_V$ the slope of a plot of U against T at constant V .

$$dU = \left(\frac{\partial U}{\partial V}\right)_T dV + C_V dT$$

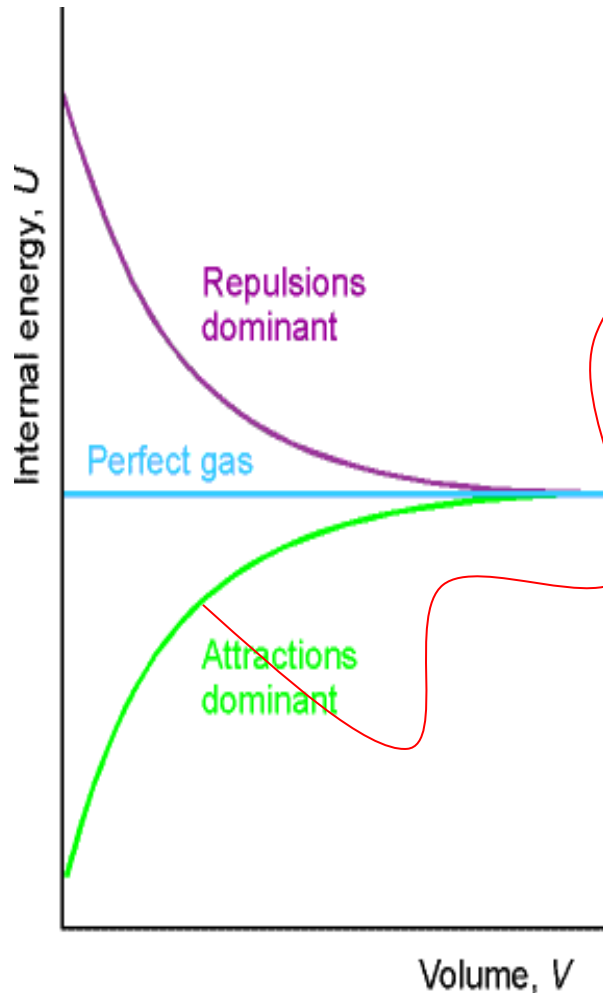
$\pi_T = \left(\frac{\partial U}{\partial V}\right)_T$, the internal pressure, the same dimension as p

$$dU = \pi_T dV + C_V dT$$

In a closed system of constant composition, dU is proportional to dV and dT , the coefficients of proportionality being the partial derivatives.



$$dU = \pi_T dV + C_V dT$$



If $dU > 0$ as the volume of the sample expands isothermal, $dV > 0$, which is the case when there are attractive forces between the particles $\pi_T > 0$

When there are no interactions between the molecules, U is independent of their separation and hence independent of the volume the sample occupies; hence $\pi_T = 0$ for a perfect gas.

Joule's Experiment

Joule's experiment was crude. In particular, the heat capacity of the apparatus was so large that the temperature change that gases do in fact cause was too small to measure. From his experiment Joule extracted an essential limiting property of a gas, a property of a perfect gas, without detecting the small deviations characteristic of real gases.

Gay-Lussac in 1807
Joule in 1843

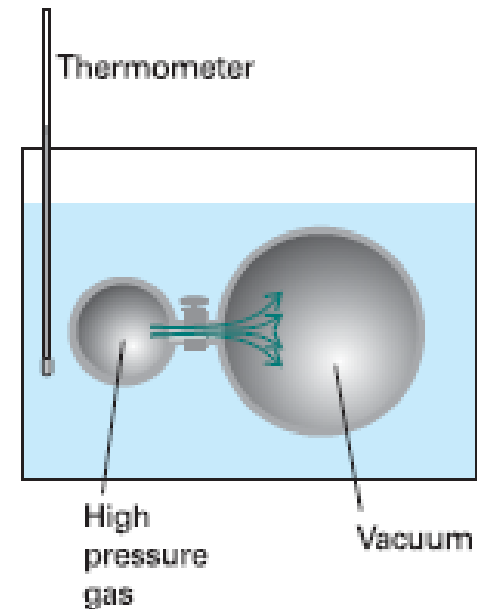


Fig. 2.26 A schematic diagram of the apparatus used by Joule in an attempt to measure the change in internal energy when a gas expands isothermally. The heat absorbed by the gas is proportional to the change in temperature of the bath.



$$\left(\frac{\partial U}{\partial V}\right)_T = 0$$

At constant temperature, changing V ,
 U not change

Discussion:

Try to justify:

$$U = U(T, p) \longrightarrow \left(\frac{\partial U}{\partial p}\right)_T = 0$$

For perfect gas

$$U = f(T)$$



$$H = U + pV$$

For perfect gases, at constant temperature:

$$\Delta H = \Delta U + \Delta(pV) = \Delta U + \Delta(nRT) = \Delta U$$

$$\left(\frac{\partial H}{\partial p} \right)_T = 0 \quad \left(\frac{\partial H}{\partial V} \right)_T = 0$$

For perfect gas

$$H = f(T)$$



$$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV$$

As for perfect gas $\left(\frac{\partial U}{\partial V}\right)_T = 0 \Rightarrow dU = \left(\frac{\partial U}{\partial T}\right)_V dT = C_V dT$

$$dH = \left(\frac{\partial H}{\partial T}\right)_p dT + \left(\frac{\partial H}{\partial p}\right)_T dp$$

As for perfect gas $\left(\frac{\partial H}{\partial p}\right)_T = 0 \Rightarrow dH = \left(\frac{\partial H}{\partial T}\right)_p dT = C_p dT$



Perfect gases, no additional work

$$\Delta U = \int C_V dT$$

Perfect gases, no additional work

$$\Delta H = \int C_p dT$$

These two equations are always established, which are not limited to constant P/V conditions.

For perfect gas

$$C_V = f(T) \quad C_p = f(T)$$



D. The relation between heat capacities

In most cases the heat capacity at constant p (C_p) of a system is larger than its heat capacity at constant V (C_V).

Most systems expand when heated at constant pressure. Such systems do work on the surroundings and therefore some of the energy supplied to them as heat escapes back to the surroundings. As a result, the temperature of the system rises less than when the heating occurs at constant volume. A smaller increase in temperature implies a larger heat capacity, so we conclude that in most cases the heat capacity at constant pressure of a system is larger than its heat capacity at constant volume.



$$C_p \quad C_V$$

$$\begin{aligned} C_p - C_V &= \left(\frac{\partial H}{\partial T}\right)_p - \left(\frac{\partial U}{\partial T}\right)_V \\ &= \left(\frac{\partial(U + PV)}{\partial T}\right)_p - \left(\frac{\partial U}{\partial T}\right)_V \\ &= \left(\frac{\partial U}{\partial T}\right)_p + p\left(\frac{\partial V}{\partial T}\right)_p - \left(\frac{\partial U}{\partial T}\right)_V \end{aligned} \quad (1)$$



$$U = U(T, V), \quad V = V(T, p)$$

$$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV$$

$$dV = \left(\frac{\partial V}{\partial T}\right)_p dT + \left(\frac{\partial V}{\partial p}\right)_T dp$$

代入 dV 表达式得：

$$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T \left[\left(\frac{\partial V}{\partial T}\right)_p dT + \left(\frac{\partial V}{\partial p}\right)_T dp \right]$$

$$= \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial p}\right)_T dp + \left[\left(\frac{\partial U}{\partial T}\right)_V + \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p \right] dT \quad (2)$$



$$dU = \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial p}\right)_T dp + \left[\left(\frac{\partial U}{\partial T}\right)_V + \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p\right] dT \quad (2)$$

$$U = f(p, T)$$

$$dU = \left(\frac{\partial U}{\partial p}\right)_T dp + \left(\frac{\partial U}{\partial T}\right)_p dT \quad (3)$$



$$\left(\frac{\partial U}{\partial T}\right)_p = \left(\frac{\partial U}{\partial T}\right)_V + \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p \quad (4)$$



$$\left(\frac{\partial U}{\partial T}\right)_p = \left(\frac{\partial U}{\partial T}\right)_V + \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p$$

(4)

$$C_p - C_V = \left(\frac{\partial U}{\partial T}\right)_p + p \left(\frac{\partial V}{\partial T}\right)_p - \left(\frac{\partial U}{\partial T}\right)_V$$

(1)

$$C_p - C_V = \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p + p \left(\frac{\partial V}{\partial T}\right)_p$$

$$= \left[p + \left(\frac{\partial U}{\partial V}\right)_T \right] \left(\frac{\partial V}{\partial T}\right)_p$$



$$C_p - C_V = \left[p + \left(\frac{\partial U}{\partial V} \right)_T \right] \left(\frac{\partial V}{\partial T} \right)_p$$

For the two heat capacities of **a perfect gas**:

$$\left(\frac{\partial U}{\partial V} \right)_T = 0, \quad \left(\frac{\partial V}{\partial T} \right)_p = \frac{nR}{p}$$

$$C_p - C_V = nR \quad C_{p,m} - C_{V,m} = R$$

It follows that the molar heat capacity of a perfect gas is about $8 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$ larger at constant pressure than at constant volume



Heat capacity of the perfect gases

$$U = f(T); \quad H = f(T)$$

$$C_V = f(T); \quad C_p = f(T)$$

$$C_{p,m} - C_{V,m} = R$$

$$\Delta U = \int C_v dT$$

$$\Delta H = \int C_p dT$$

这两个公式不受限是否等容、等压

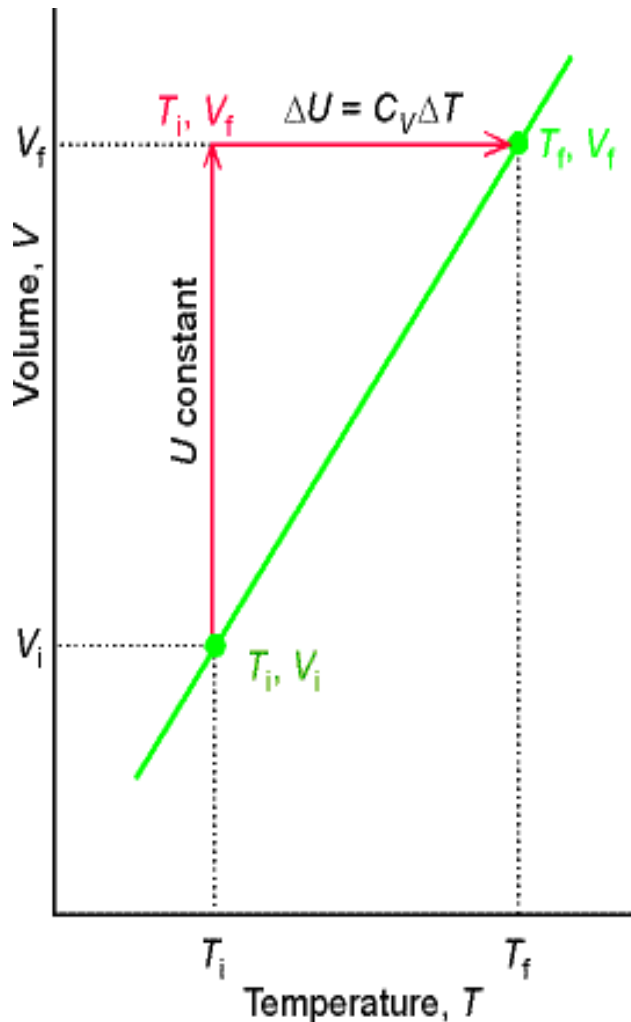
2.6 Adiabatic changes

$$q=0, \Delta U=w$$

p, V, T changing

We are now equipped to deal with the changes that occur when a perfect gas expands adiabatically. A decrease in temperature should be expected: because work is done but no heat enters the system, the internal energy falls, and therefore the temperature of the working gas also falls. In molecular terms, the kinetic energy of the molecules falls as work is done, so their average speed decreases, and hence the temperature falls.

A. The work of adiabatic change



The change in internal energy:

$$\Delta U = C_V (T_f - T_i) = C_V \Delta T$$

For adiabatic expansion

$$q=0, \Rightarrow \Delta U = w_{\text{ad}}$$

$$w_{\text{ad}} = C_V \Delta T$$

“ad” denotes an adiabatic process



Consider a stage in the expansion when the pressure inside and out is p (reversible expansion):

$$dw = -pdV$$

For a perfect gas:

$$dU = dw = C_V dT \quad \therefore C_V dT = -pdV$$

$$\frac{C_V dT}{T} = -\frac{nR dV}{V}$$

$$C_V \int_{T_i}^{T_f} \frac{dT}{T} = -nR \int_{V_i}^{V_f} \frac{dV}{V}$$

$$C_V \ln \frac{T_f}{T_i} = -nR \ln \frac{V_f}{V_i}$$



$$C_V \ln \frac{T_f}{T_i} = -nR \ln \frac{V_f}{V_i}$$

$$\ln \left(\frac{T_f}{T_i} \right)^{C_V/nR} = \ln \frac{V_i}{V_f}$$

$$\text{If, } c = \frac{C_V}{nR} = C_{V,m}/R$$

$$\ln \left(\frac{T_f}{T_i} \right)^c = \ln \frac{V_i}{V_f}$$

$$T_f = T_i \left(\frac{V_i}{V_f} \right)^{1/c}$$

$$V_f T_f^c = V_i T_i^c$$

For an adiabatic system doing reversible expansion work



B. Heat capacity ratio and adiabats

$$\frac{p_i V_i}{p_f V_f} = \frac{T_i}{T_f}$$

$$\frac{T_i}{T_f} = \left(\frac{V_f}{V_i} \right)^{1/c} = \left(\frac{V_f}{V_i} \right)^{\gamma-1}$$

$$c = C_{V,m}/R$$

$$\begin{aligned} \gamma &= 1/c + 1 \\ &= \frac{C_{V,m} + R}{C_{V,m}} = \frac{C_{p,m}}{C_{V,m}} \end{aligned}$$

$$p_i V_i^\gamma = p_f V_f^\gamma$$



The change in pressure resulting from an adiabatic, reversible expansion of a perfect gas:

$$pV^\gamma = \text{constant}$$

The **heat capacity ratio** is defined as:

$$\gamma = \frac{C_{p,m}}{C_{V,m}}$$



Since the heat capacity at constant pressure is greater than the heat capacity at constant volume

$$\gamma > 1$$

For a perfect gas, it follows that:

$$\gamma = \frac{C_{V,m} + R}{C_{V,m}}$$

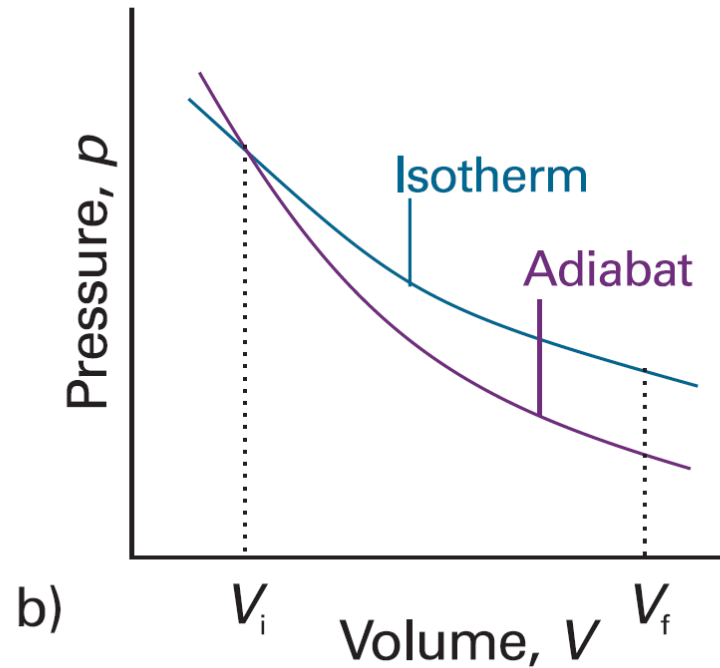
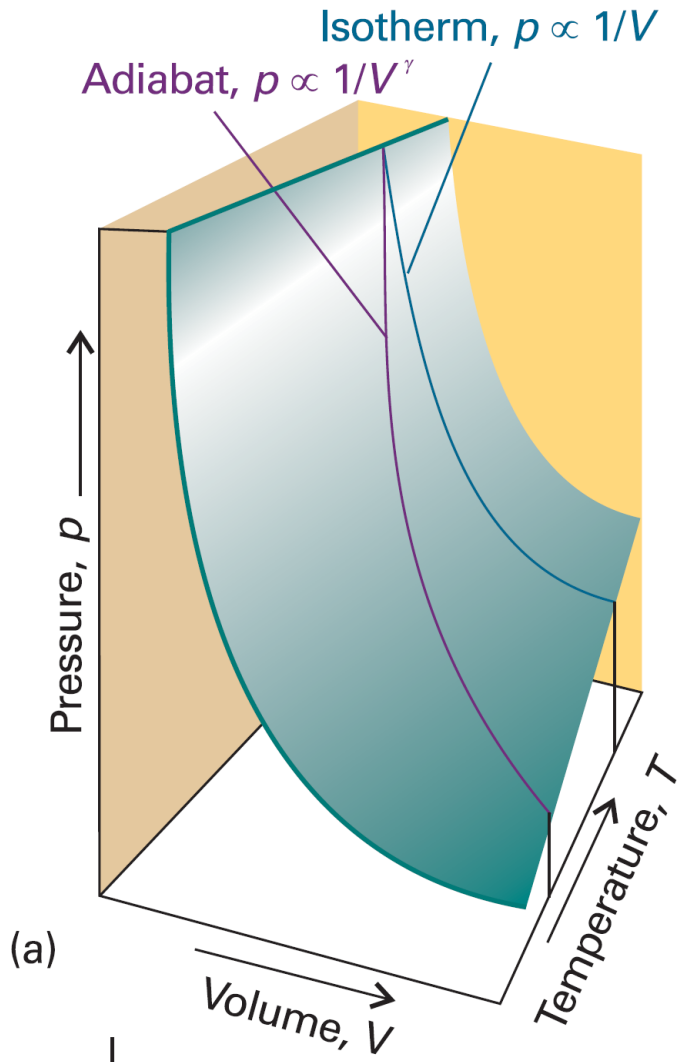
Example

When a sample of argon (for which $\gamma = 5/3$) at 100 kPa expands reversibly and adiabatically to twice its initial volume, the final pressure will be:

$$pV^\gamma = \text{constant}$$

$$\begin{aligned} p_f &= p_i \left(\frac{V_i}{V_f} \right)^\gamma \\ &= (100 \text{ kPa}) \times \left(\frac{1}{2} \right)^{5/3} \\ &= 31.5 \text{ kPa} \end{aligned}$$

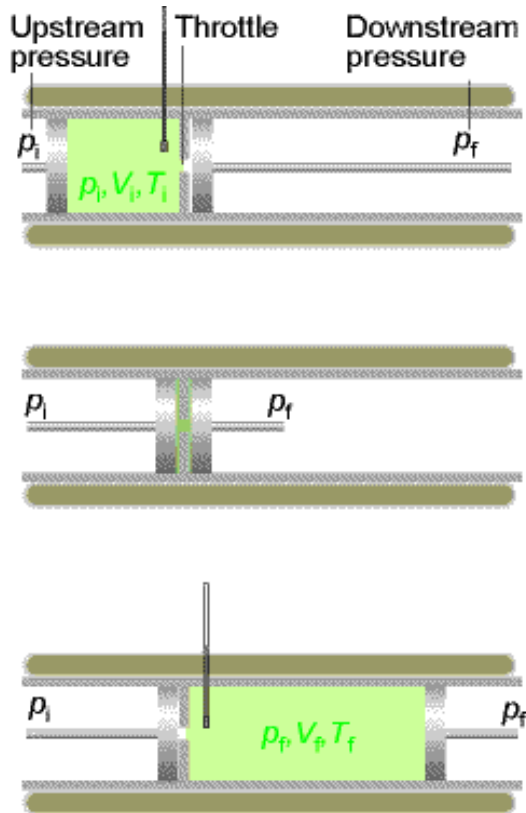
Adiabats (绝热线)



An adiabat depicts the variation of pressure with volume when a gas expands adiabatically. (a) An adiabat for a perfect gas undergoing reversible expansion. (b) Note that the pressure declines more steeply for an adiabat than it does for an isotherm because the temperature decreases in the former. 86



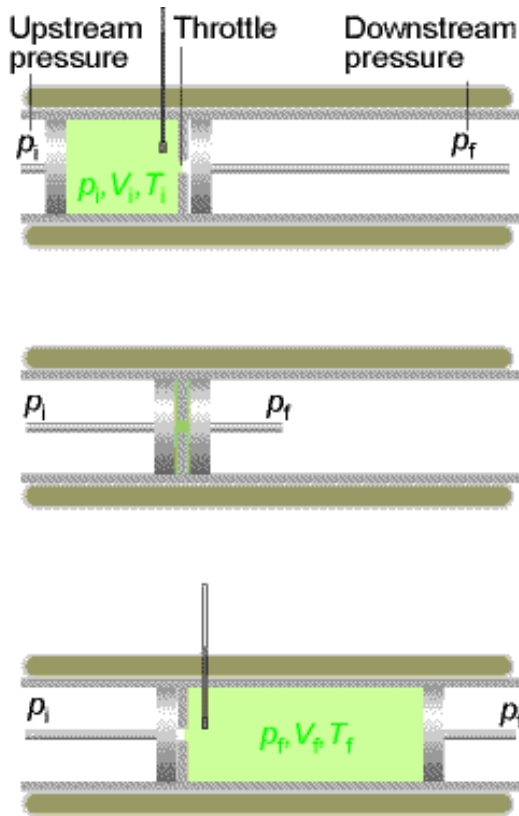
2.6 The first law for real gases-Joule-Thomson effect



The thermodynamic basis of Joule-Thomson expansion.

Adiabatic expansion:

let a gas expand through a porous barrier from one constant pressure to another, and monitored the difference of T that arose from the expansion. The process was adiabatic. It was observed a lower T on the low-pressure side, the difference in T being proportional to the p difference maintained. This cooling by adiabatic expansion is now called the **Joule-Thomson effect**.



The thermodynamic basis of Joule-Thomson expansion.

Because all changes to the gas occur adiabatically: $q = 0$.

The gas on the left is compressed and the work done:

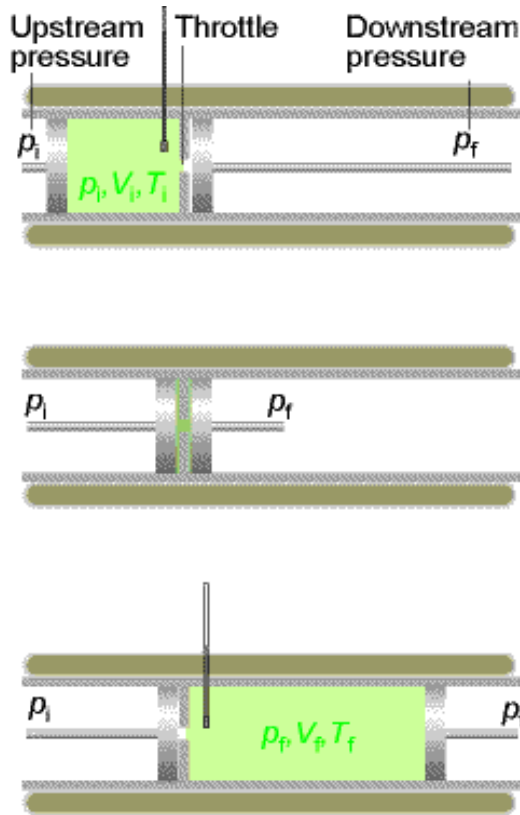
$$w_L = - p_i (0 - V_i) = p_i V_i$$

The work done in right:

$$w_R = - p_f (V_f - 0) = -p_f V_f$$

The total work done:

$$w = w_L + w_R = p_i V_i - p_f V_f$$



The thermodynamic basis of Joule-Thomson expansion.

Since $q = 0$, then $U = w + q = w$.
the change of U of the gas as it moves from one side of the throttle to the other is:

$$U_f - U_i = w = p_i V_i - p_f V_f$$

$$U_f + p_f V_f = U_i + p_i V_i$$

$$H_f = H_i$$

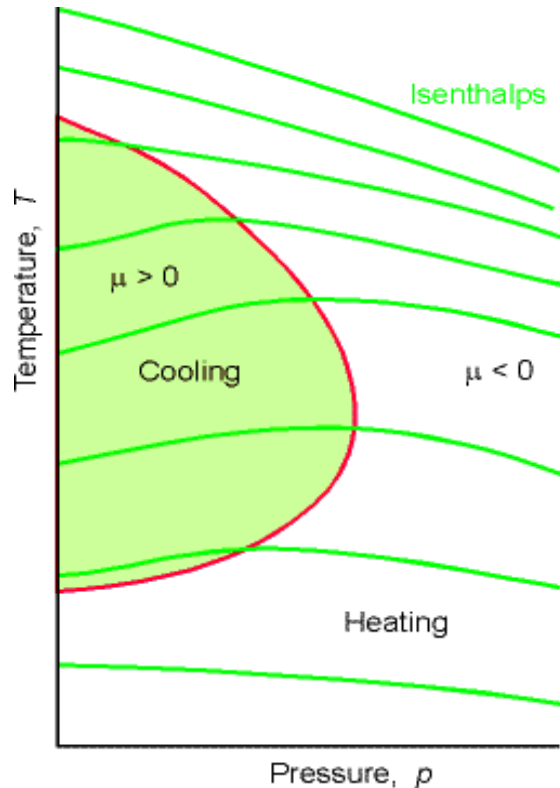
The expansion occurs without change of enthalpy: an isenthalpic process.



The analysis of the Joule-Thomson coefficient is central to the technological problems associated with the liquefaction of gases.

The property is the ratio of the T change to the change of p , $\Delta T/\Delta p$. Adding the constraint of constant enthalpy and taking the limit of small Δp implies the quantity measured is $(dT/dp)_H$, which is the **Joule-Thomson coefficient**, μ . And μ is the ratio of the change in temperature to the change in pressure when a gas expands under adiabatic conditions.

$$\mu = \left(\partial T / \partial p \right)_H$$



The sign of the Joule-Thomson coefficient, μ

For a perfect gas, $\mu = 0$. But Real gases have nonzero μ and, depending on the identity of the gas, p and T , the relative magnitudes of the attractive and repulsive intermolecular forces, the sign of the coefficient may be either positive or negative.

A **positive sign** implies that dT is negative when dp is negative, in which case the gas **cools** on expansion. A **negative sign** implies **heating** process.

$$H = H(T, p)$$

$$dH = \left(\frac{\partial H}{\partial T} \right)_p dT + \left(\frac{\partial H}{\partial p} \right)_T dp \quad dH = 0$$

$$\left(\frac{\partial T}{\partial p} \right)_H = - \frac{\left(\frac{\partial H}{\partial p} \right)_T}{\left(\frac{\partial H}{\partial T} \right)_p}$$

$$\mu_{J-T} = \left(\frac{\partial T}{\partial p} \right)_H = - \frac{\left[\frac{\partial(U + pV)}{\partial p} \right]_T}{C_p}$$

$$= \left\{ - \frac{1}{C_p} \left(\frac{\partial U}{\partial p} \right)_T \right\} + \left\{ - \frac{1}{C_p} \left(\frac{\partial(pV)}{\partial p} \right)_T \right\}$$

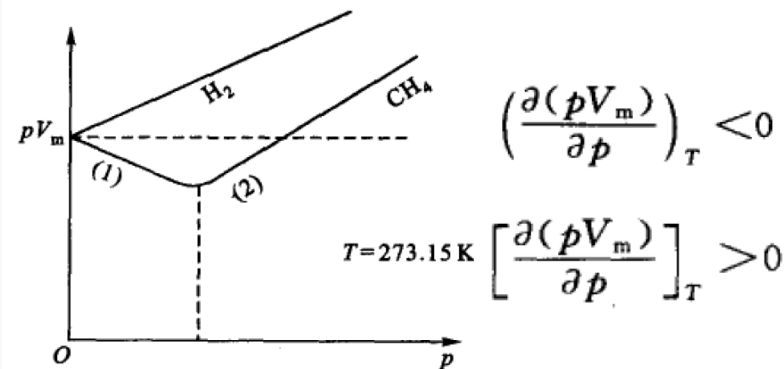
The perfect gases

$$\left(\frac{\partial U}{\partial p} \right)_T = 0, \quad \left[\frac{\partial(pV)}{\partial p} \right]_T = 0$$

Real gases

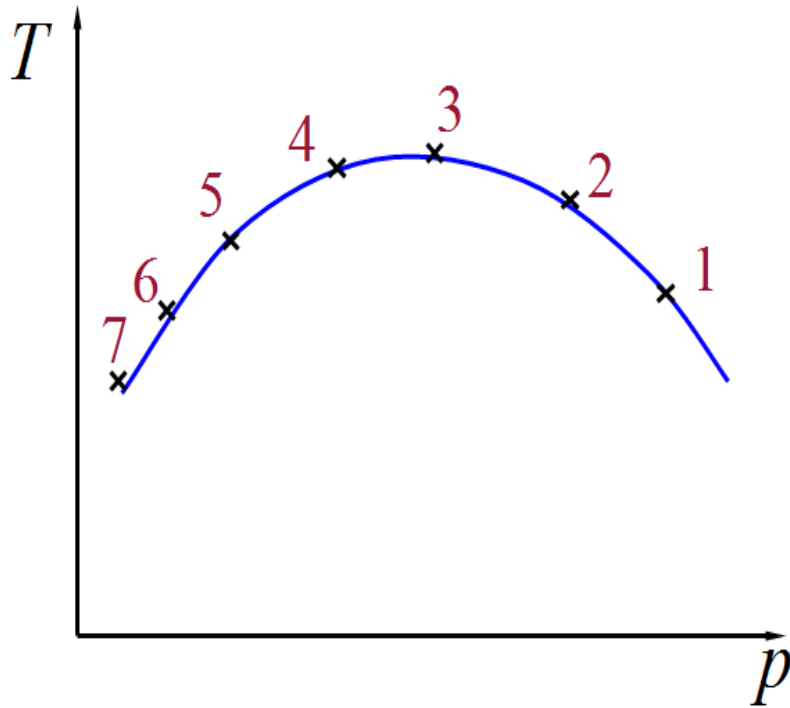
$$\left(\frac{\partial U}{\partial p} \right)_T < 0, \quad C_p > 0,$$

$$\left\{ - \frac{1}{C_p} \left(\frac{\partial U}{\partial p} \right)_T \right\} > 0$$

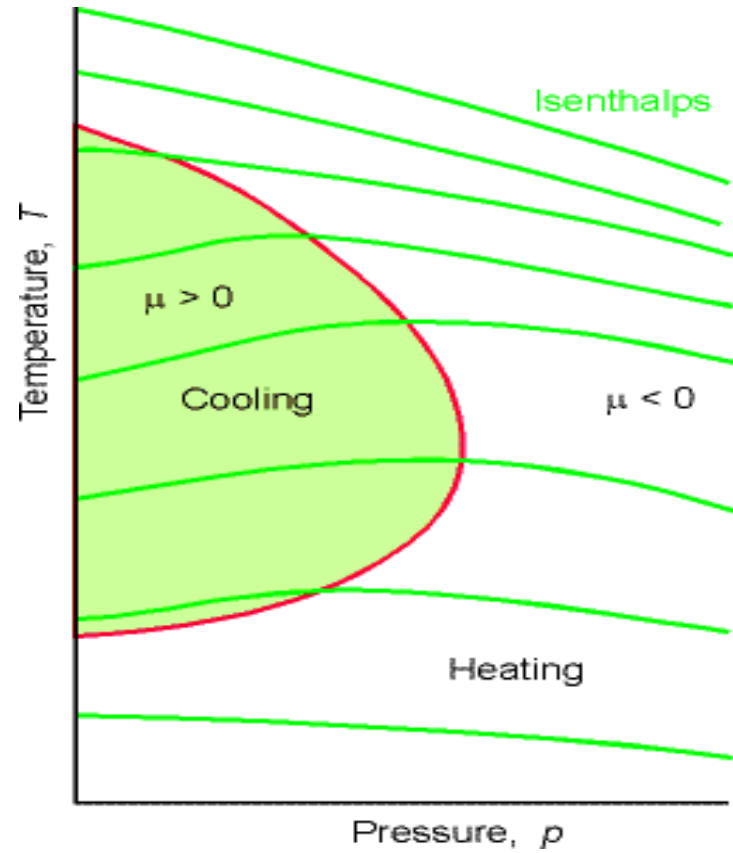




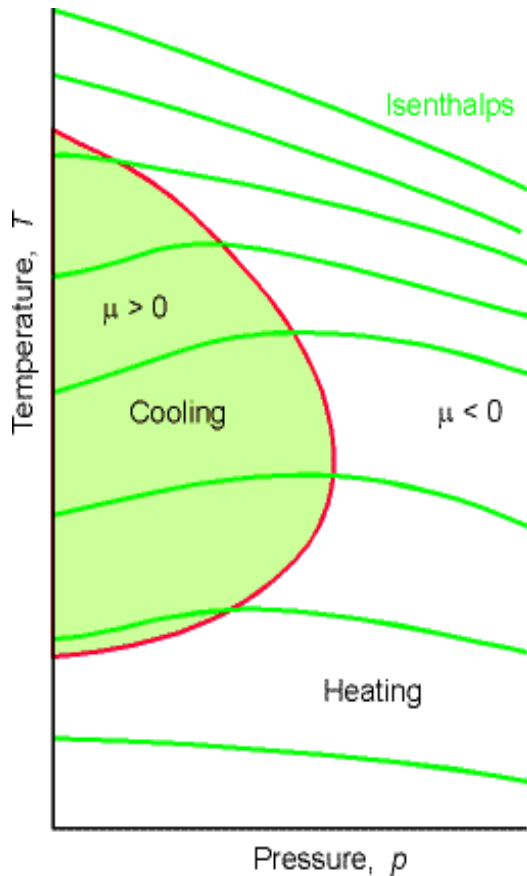
Isenthalps & inversion curve



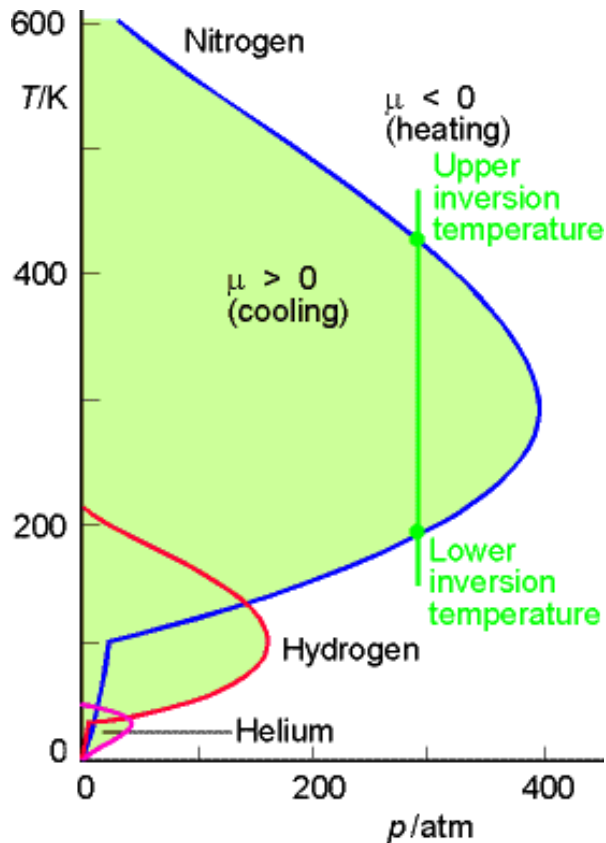
等焓线



转化线



Inside the boundary, $\mu > 0$ and outside it is negative. The T corresponding to the boundary at a given p is the **inversion T** of the gas at that p . Reduction of p under adiabatic conditions moves the system along one of the **isenthalps**. The inversion T curve runs through the points of the isenthalps where their slopes change from negative to positive.



Gases that show a heating effect ($\mu < 0$) at one T show a cooling effect ($\mu > 0$) when the T is below their upper inversion temperature, T_I . A gas typically has two inversion temperatures, one at high temperature and the other at low.

The sign of the Joule-Thomson coefficient, μ

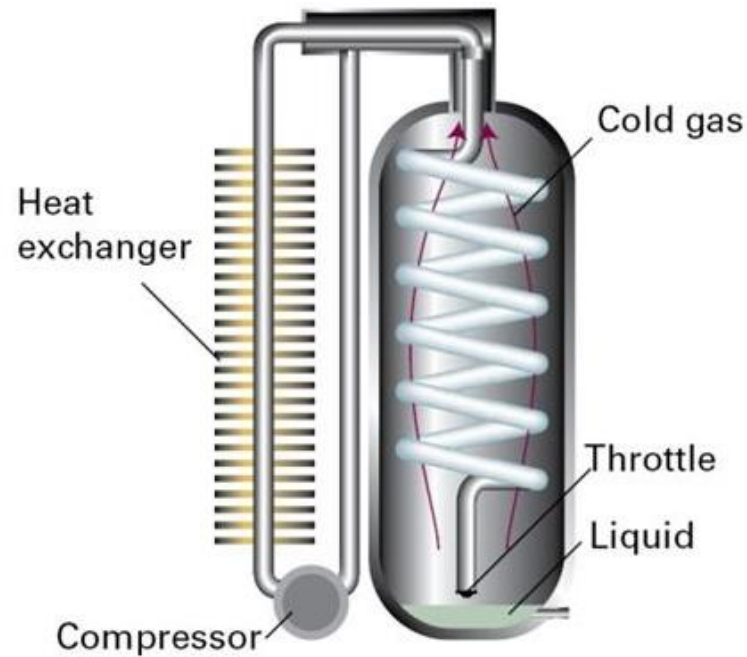


Figure 2D.12 The principle of the Linde refrigerator is shown in this diagram. The gas is recirculated, and so long as it is beneath its inversion temperature it cools on expansion through the throttle. The cooled gas cools the high-pressure gas, which cools still further as it expands. Eventually liquefied gas drips from the throttle.





Thermochemistry: a branch of thermodynamics, concentrating on the heat produced or required by chemical reactions.

Process releasing heat—exothermic:

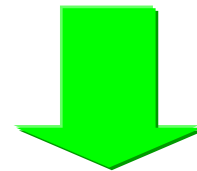
$$\Delta H < 0.$$

Process absorbing heat—endothermic:

$$\Delta H > 0.$$

2.7 Standard enthalpy changes

The standard enthalpy change, $\Delta H^\ominus$, the change in enthalpy for a process in which the initial and final substances are in their standard states.



The standard state: of the pure form a substance at 1 bar and at a specified temperature.

Conventionally, $T=298.15\text{ K}$ ($25\text{ }^\circ\text{C}$)

For example:

The standard state of liquid ethanol at 298 K is pure liquid ethanol at 298 K and 1 bar.

The standard state of solid iron at 500 K is pure iron at 500 K and 1 bar.

The standard enthalpy change for a reaction or a physical process is the difference between the products in their standard states and the reactants in their standard states, all at the same specified temperature.

A. Enthalpies of physical change

The standard enthalpy of transition, $\Delta_{\text{trs}}H^\ominus$: the standard enthalpy change accompanying a change of physical state.

The standard enthalpy of vaporization - enthalpy change per mole when a pure liquid at 1 bar vaporizes to a gas at 1 bar:

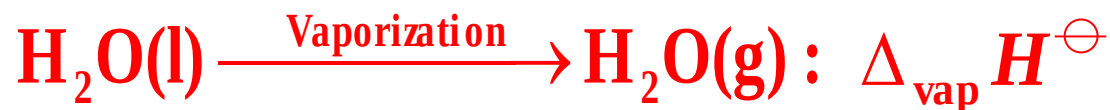
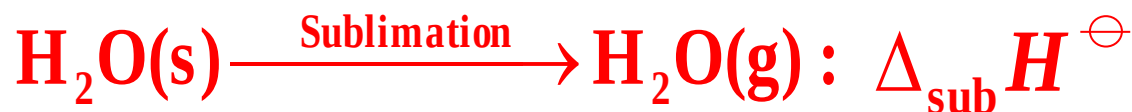
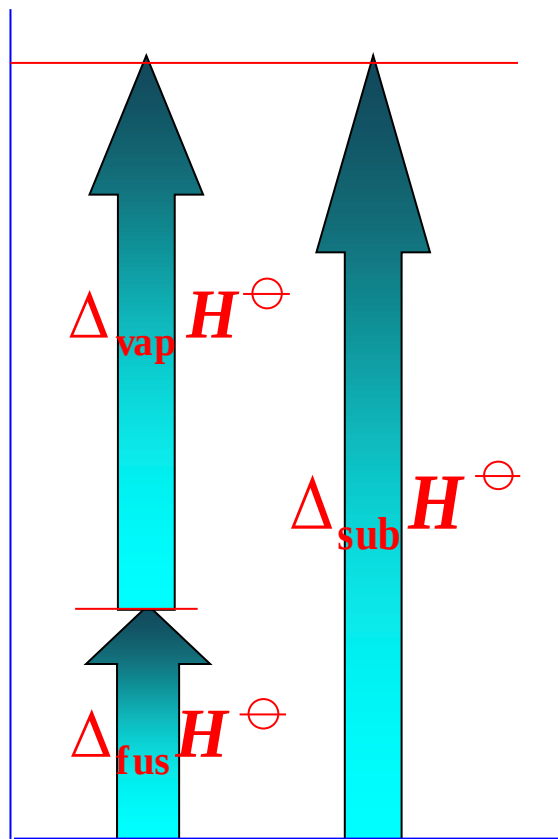


The standard enthalpy of fusion - enthalpy change accompanying the conversion of a solid to a liquid

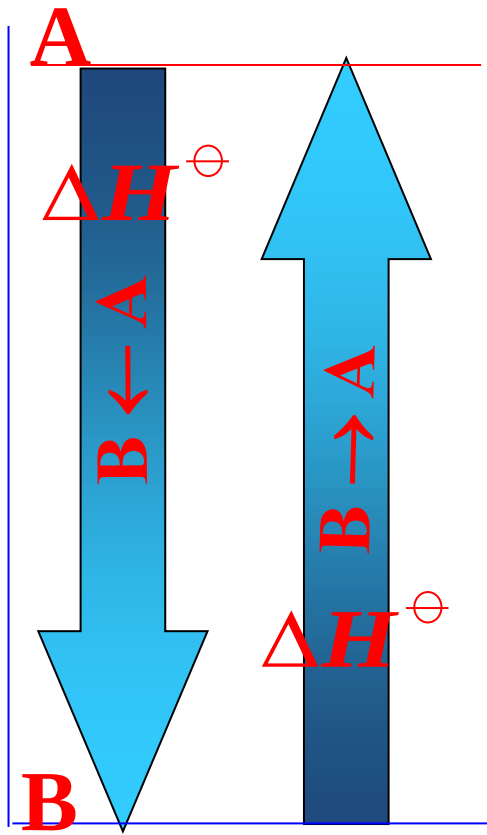




Because enthalpy is a state function, a change in enthalpy is independent of the path between the two states.



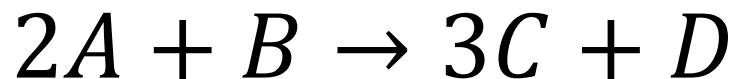
$$\Delta_{\text{sub}} H^\ominus = \Delta_{\text{fus}} H^\ominus + \Delta_{\text{vap}} H^\ominus$$



$$\Delta H^{\ominus}(\text{A} \rightarrow \text{B}) = -\Delta H^{\ominus}(\text{B} \rightarrow \text{A})$$

B. Enthalpies of chemical change

The standard reaction enthalpy, $\Delta_r H^\ominus$: the change in enthalpy when reactants in their standard states change to products in their standard states.



$$\begin{aligned}\Delta_r H^\ominus &= \{3H_m^\ominus(C) + H_m^\ominus(D)\} - \{2H_m^\ominus(A) + H_m^\ominus(B)\} \\ &= \sum v_J H_m^\ominus(J)\end{aligned}$$

$H_m^\ominus(J)$: the standard molar enthalpy of species J at the temperature of interest.

v_J : the stoichiometric coefficients



C. Hess's law

The standard enthalpy of an overall reaction is the sum of the standard enthalpies of the individual reactions into which a reaction may be divided.



The standard reaction enthalpy for the hydrogenation of propene:



$$\Delta_r H^\ominus = -124 \text{ kJ mol}^{-1}$$

The standard reaction enthalpy for combustion of propane:



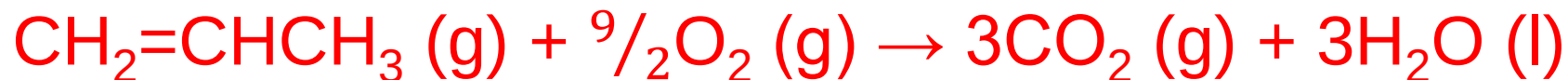
$$\Delta_r H^\ominus = -2220 \text{ kJ mol}^{-1}$$

The standard reaction enthalpy for the formation of water:



$$\Delta_r H^\ominus = -286 \text{ kJ mol}^{-1}$$

The standard reaction enthalpy for combustion of propene?



+	$\text{CH}_2=\text{CHCH}_3 (\text{g}) + \text{H}_2 (\text{g}) \rightarrow \text{CH}_3\text{CH}_2\text{CH}_3 (\text{g})$	-124
+	$\text{CH}_3\text{CH}_2\text{CH}_3 (\text{g}) + 5\text{O}_2 (\text{g}) \rightarrow 3\text{CO}_2 (\text{g}) + 4\text{H}_2\text{O} (\text{l})$	-2220
-	$\text{H}_2 (\text{g}) + \frac{1}{2}\text{O}_2 (\text{g}) \rightarrow \text{H}_2\text{O} (\text{l})$	-286
	$\text{CH}_2=\text{CHCH}_3 (\text{g}) + \frac{9}{2}\text{O}_2 (\text{g}) \rightarrow 3\text{CO}_2 (\text{g}) + 3\text{H}_2\text{O} (\text{l})$	-2058 KJ mol ⁻¹



1. Standard enthalpies of formation

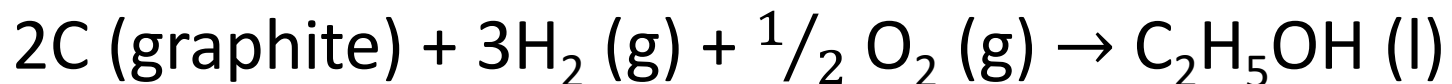
- **The standard enthalpy of formation, $\Delta_f H^\ominus$:** the standard reaction enthalpy for the formation of the compound from its elements in their reference states.
- **The reference state** of an element is its most stable state at the **specified temperature and 1 bar**.
- The standard enthalpy of formation for an element in its reference state is **zero**.



The standard enthalpy of formation for ethanol (C₂H₅OH):

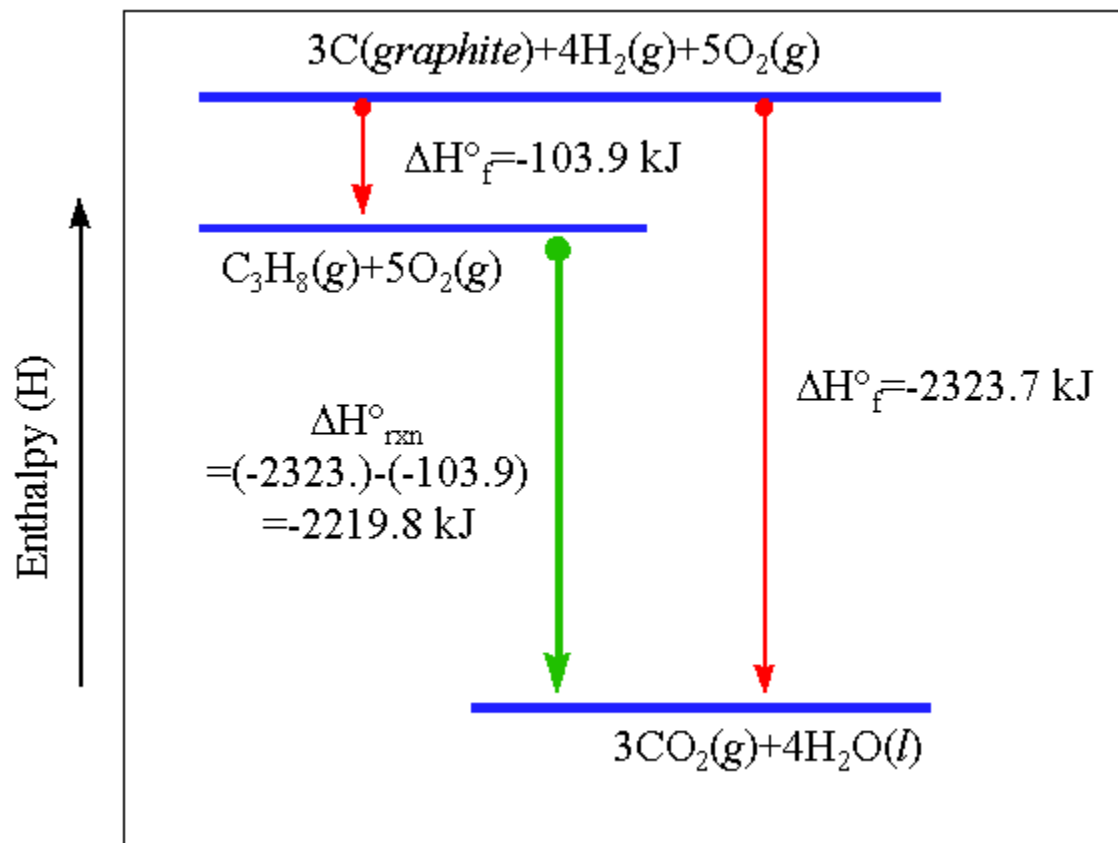
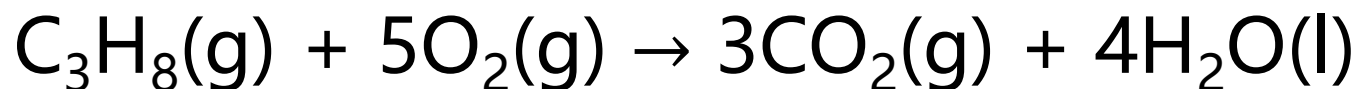
Table 2C.4

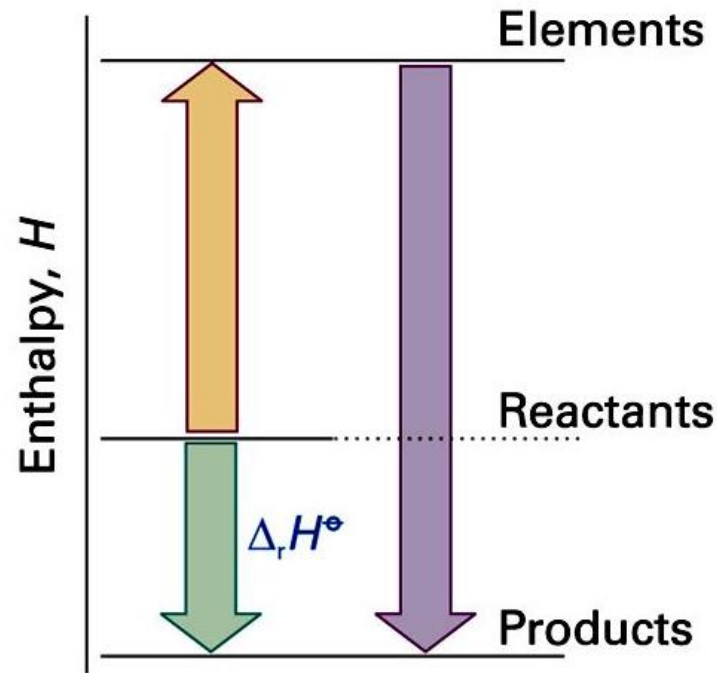
$$\Delta_f H^\ominus(298\text{ K}) = -277.69\text{ kJ mol}^{-1}$$



- Elemental source of oxygen is O₂ and not O because O₂ is the stable form of oxygen at 25 °C and 1 bar, likewise with H₂;
- Elemental source of carbon is specified as graphite (and not, for example, diamond) because graphite is the lowest energy form of carbon at room temp and 1 bar;
- Why is the O₂ stoichiometry left at "1/2"? The stoichiometry of formation reactions always indicates the formation of 1 mol of product. Thus, $\Delta_f H^\ominus$ values are reported as kJ / mole of the substance produced.

2. Standard enthalpies of reaction





$$\Delta_r H^\ominus = \sum_{\text{Products}} v \Delta_f H^\ominus - \sum_{\text{Reactants}} v \Delta_f H^\ominus$$

$$\Delta_r H^\ominus = \sum_J v_J H_f^\ominus$$



Calculate the standard enthalpy of the reaction:

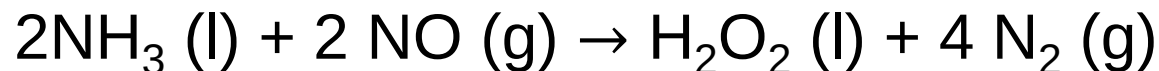


Table 2C.4

$$\Delta_f H^\ominus (\text{NH}_3, \text{l}) = 264.0 \text{ kJ mol}^{-1}$$

$$\Delta_f H^\ominus (\text{NO}, \text{g}) = -90.25 \text{ kJ mol}^{-1}$$

$$\Delta_f H^\ominus (\text{H}_2\text{O}_2, \text{l}) = -187.78 \text{ kJ mol}^{-1}$$

$$\Delta_f H^\ominus (\text{N}_2, \text{g}) = 0 \text{ kJ mol}^{-1}$$

$$\Delta_r H^\ominus = \sum_{\text{Products}} \nu \Delta_f H^\ominus - \sum_{\text{Reactants}} \nu \Delta_f H^\ominus$$

$$= \{ \Delta_f H^\ominus (\text{H}_2\text{O}_2, \text{l}) + 4 \times \Delta_f H^\ominus (\text{N}_2, \text{g}) \} - \{ 2 \times \Delta_f H^\ominus (\text{NH}_3, \text{l}) + 2 \times \Delta_f H^\ominus (\text{NO}, \text{g}) \}$$

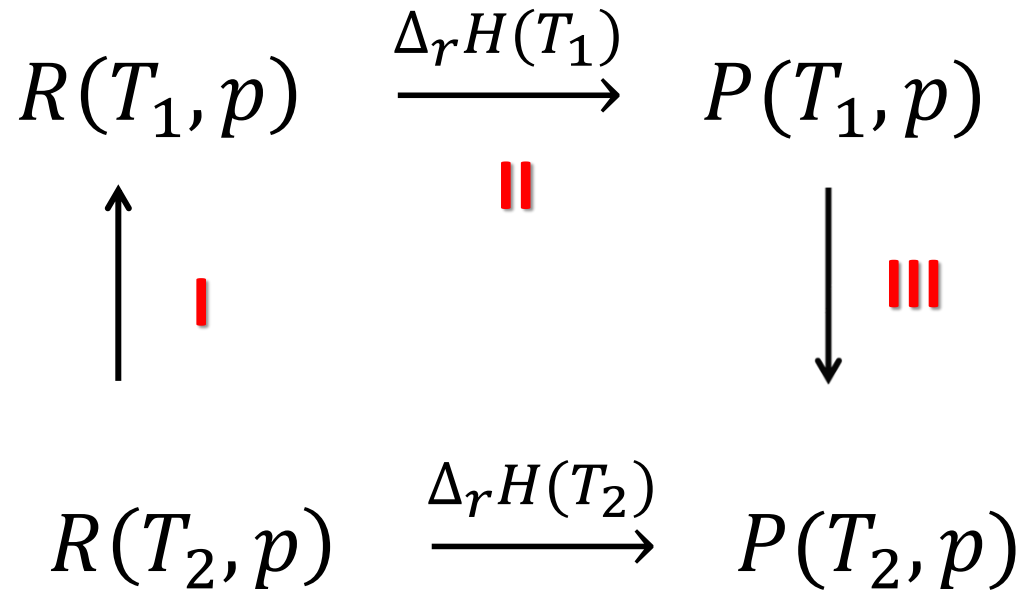
$$= -896.3 \text{ kJ mol}^{-1}$$

3. The temperature dependence of reaction enthalpies Kirchhoff's law

$$\Delta_r H^\ominus(T_2) = \Delta_r H^\ominus(T_1) + \int_{T_1}^{T_2} \Delta_r C_p^\ominus dT$$

$$\Delta_r C_p^\ominus = \sum_{\text{Products}} \nu C_{p,m}^\ominus - \sum_{\text{Reactants}} \nu C_{p,m}^\ominus$$

Where $\Delta_r C_p^\ominus$ is the difference of the molar heat capacities of products and reactants under standard conditions weighted by the stoichiometric coefficients.



$$\begin{aligned} \Delta_r H(T_2) &= \Delta H_I + \Delta_r H(T_1) + \Delta H_{III} \\ &= \int_{T_2}^{T_1} C_{p,m}(R) dT + \Delta_r H(T_1) + \int_{T_1}^{T_2} C_{p,m}(P) dT \end{aligned}$$

$$\Delta_r H(T_2) = \Delta_r H(T_1) + \int_{T_1}^{T_2} \Delta_r C_{p,m} dT$$

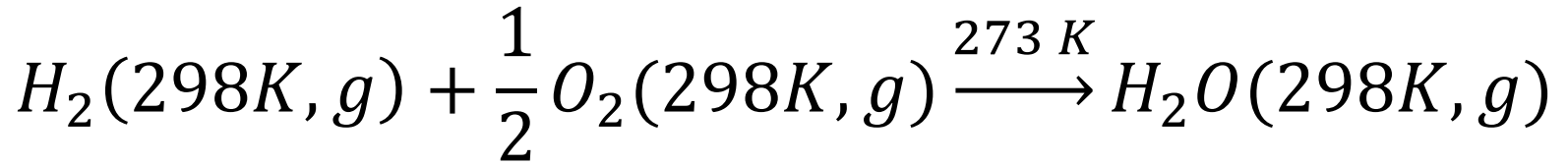


The standard enthalpy of formation of gaseous H_2O at 298 K is $-241.82 \text{ kJ mol}^{-1}$. Estimate its value at 100°C given the following values of the molar heat capacities at constant pressure: $\text{H}_2\text{O(g)}$: $33.58 \text{ J K}^{-1} \text{ mol}^{-1}$; $\text{H}_2\text{(g)}$: $28.84 \text{ J K}^{-1} \text{ mol}^{-1}$; $\text{O}_2\text{(g)}$: $29.37 \text{ J K}^{-1} \text{ mol}^{-1}$. Assume that the heat capacities are independent of temperature.

$$\Delta_r H^\ominus(T_2) = \Delta_r H^\ominus(T_1) + \int_{T_1}^{T_2} \Delta_r C_p^\ominus dT$$

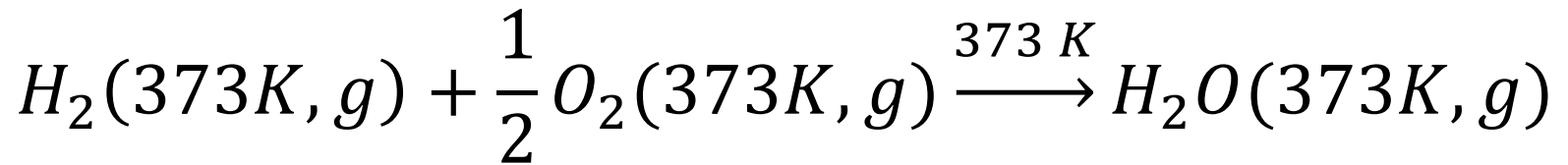
$$\Delta_r H^\ominus(T_2) = \Delta_r H^\ominus(T_1) + (T_2 - T_1) \Delta_r C_p^\ominus$$

$$\Delta_r H^\ominus(298\text{ K}) = -241.82\text{ kJ} \cdot \text{mol}^{-1}$$



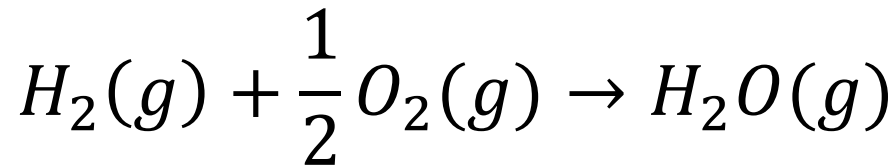
ΔH_1 = reactants change temperature

ΔH_2 = products change temperature



$$\Delta_r H^\ominus(373\text{ K}) = ?$$

$$\Delta_r H(373\text{ K}) = \Delta H_1 + \Delta_r H(273\text{ K}) + \Delta H_2$$



$$\Delta_r C_p^\ominus = C_p^\ominus(H_2O, g) - \left\{ C_p^\ominus(H_2, g) + \frac{1}{2} C_p^\ominus(O_2, g) \right\}$$

$$\begin{aligned} \Delta_r C_p^\ominus &= 33.58 J \cdot K^{-1} \cdot mol^{-1} - (28.84 J \cdot K^{-1} \cdot mol^{-1} + \frac{1}{2} \times 29.37 J \cdot K^{-1} \cdot mol^{-1}) \\ &= -9.94 J \cdot K^{-1} \cdot mol^{-1} \end{aligned}$$

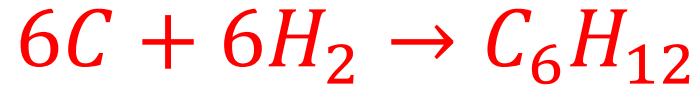
$$\Delta_r H^\ominus(T_2) = \Delta_r H^\ominus(T_1) + (T_2 - T_1) \Delta_r C_p^\ominus$$

$$\begin{aligned} \Delta_r H^\ominus(373) &= -241.82 kJ \cdot mol^{-1} + 75 K \times (-9.94 \times 10^{-3} kJ \cdot K^{-1} \cdot mol^{-1}) \\ &= -242.6 kJ \cdot mol^{-1} \end{aligned}$$



Self-test2.9

Estimate the standard enthalpy of formation of cyclohexane at 400K from the data in Table 2.5 at the end of book.

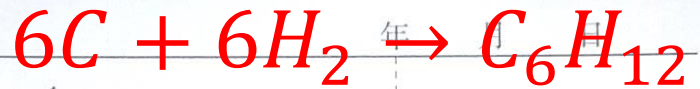


$$\Delta_r H^\ominus(T_2) = \Delta_r H^\ominus(T_1) + \int_{T_1}^{T_2} \Delta_r C_p^\ominus dT$$

$$\Delta_f H^\ominus(T_2) = \Delta_f H^\ominus(298K) + \int_{298}^{T_2} \Delta_r C_p^\ominus dT$$



For C_6H_{12}



$$\Delta_f H^\ominus(298K) = -156 \text{ kJ/mol}$$

$$C_{p,m}^\ominus(C) = 8.527 \text{ J/K}\cdot\text{mol} \quad C_{p,m}^\ominus(H_2) = 28.824 \text{ J/K}\cdot\text{mol}$$

$$C_{p,m}^\ominus(C_6H_{12}) = 156.5 \text{ J/K}\cdot\text{mol}$$

$$i) \Delta_r C_p^\ominus = C_{p,m}^\ominus(C_6H_{12}) - \{6 \times C_{p,m}^\ominus(C) + 6 \times C_{p,m}^\ominus(H_2)\}$$

$$= 156.5 \text{ J/K}\cdot\text{mol} - (6 \times 8.527 + 6 \times 28.824) \text{ J/K}\cdot\text{mol}$$

$$= -67.606 \text{ J/K}\cdot\text{mol}$$

$$\Delta_f H^\ominus(400K) = \Delta_f H^\ominus(298K) + \int_{298}^{400} -67.606 \times 10^{-3} dT$$

$$= -156 \text{ kJ/mol} - (6.76 \times 10^{-2} \times 102) \text{ kJ/mol}$$

$$\Delta_f H^\ominus(400K) = -162.9 \text{ kJ/mol}$$



Conditions which influence enthalpy changes include:

- temperature
- state of reactants and products (s, g, l, aq)



3. Carbon dioxide (CO₂) and H₂O can be produced from the reaction between CH₄ and O₂. Answer the following questions using data given below.

1) Calculate the standard enthalpy of the reaction at 298K under 1 bar (6 points).

2) Figure out the molar reaction enthalpy at the temperature of 500 K. (9 points)

$$\Delta_f H^\ominus(\text{CO}, \text{g}) = -110.53 \text{ kJ mol}^{-1}$$

$$\Delta_f H^\ominus(\text{CO}_2, \text{g}) = -393.51 \text{ kJ mol}^{-1}$$

$$\Delta_f H^\ominus(\text{CH}_4, \text{g}) = -74.81 \text{ kJ mol}^{-1}$$

$$\Delta_f H^\ominus(\text{H}_2\text{O}, \text{l}) = -285.83 \text{ kJ mol}^{-1}$$

$$C_{p,m}(\text{CH}_4, \text{g}) = 35.31 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$C_{p,m}(\text{CO}_2, \text{g}) = 37.11 \text{ J K}^{-1} \text{ mol}^{-1}$$

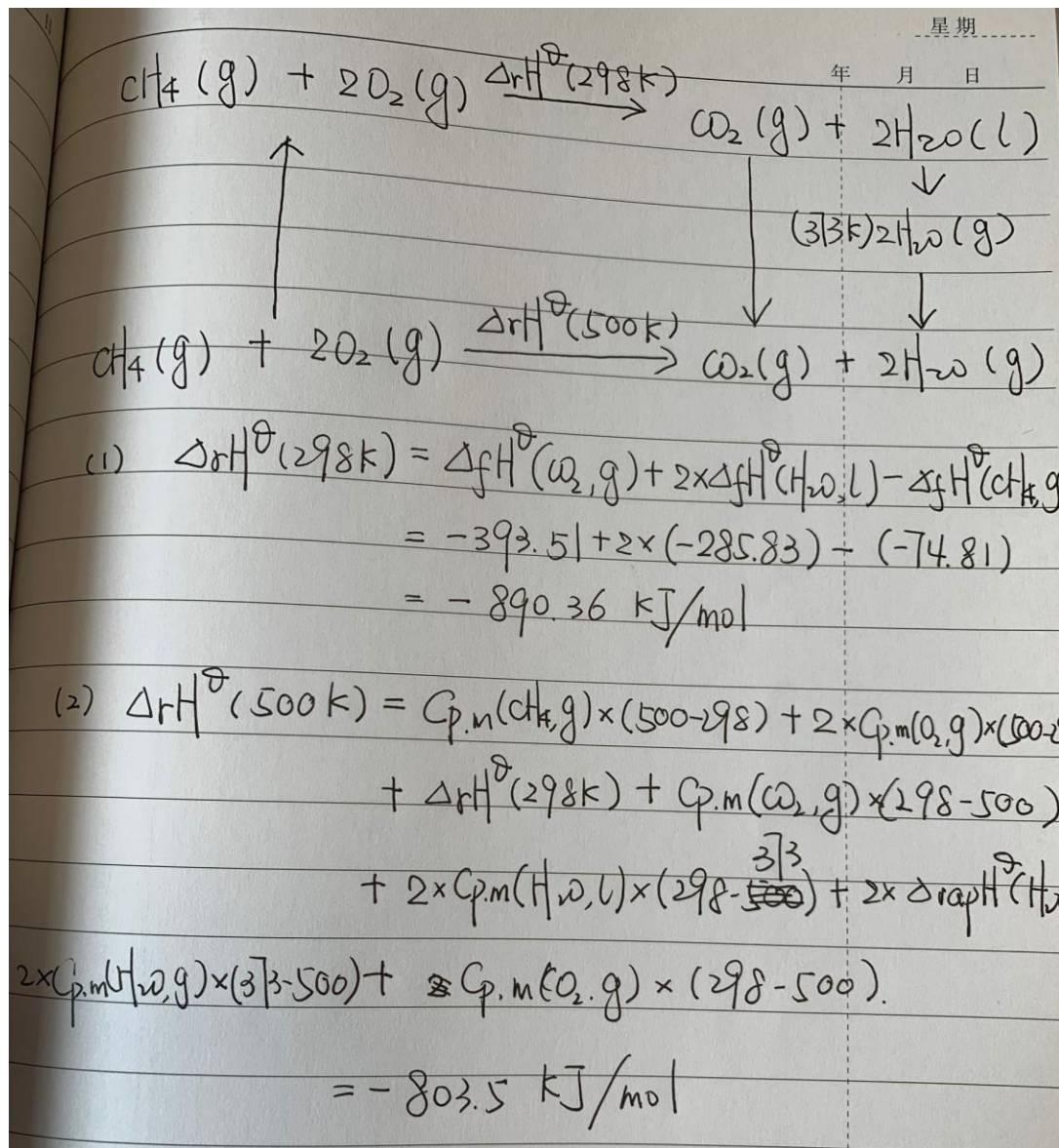
$$C_{p,m}(\text{O}_2, \text{g}) = 36.16 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$C_{p,m}(\text{CO}, \text{g}) = 29.14 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$C_{p,m}(\text{H}_2\text{O}, \text{l}) = 75.291 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$C_{p,m}(\text{H}_2\text{O}, \text{g}) = 33.58 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$\Delta_{\text{vap}} H^\ominus(\text{H}_2\text{O}) = 40.656 \text{ kJ mol}^{-1}$$





Summary

The energies

- The internal energy, U

The first law of thermodynamics

$$\Delta U = q + w$$



- Work (w)

Free expansion

$$w = 0$$

At a constant pressure

$$w = -p_{ex}\Delta V$$

Reversible

$$w = -\int_{V_i}^{V_f} p dV$$

Isothermal reversible

$$w = -nRT \ln \frac{V_f}{V_i}$$

- Heat (q)

$$q_V = \Delta U \quad \text{恒容}$$

$$q_p = \Delta H \quad \text{恒压}$$



Heat capacity of the perfect gases

$$C = \frac{\partial q}{\partial T}$$



$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

$$C_p = \left(\frac{\partial H}{\partial T}\right)_p$$

$$\Delta U = \int C_v dT$$

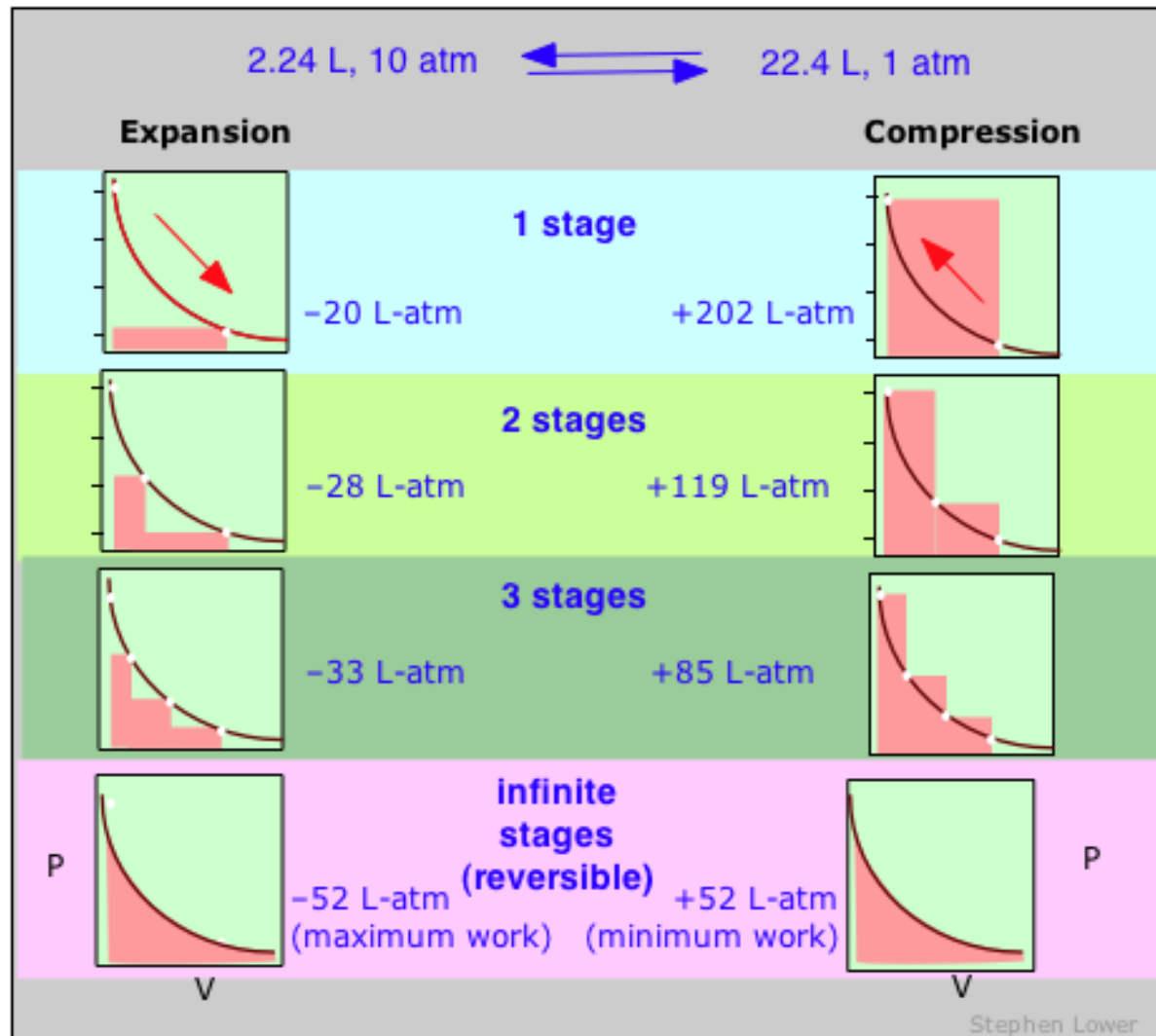
$$\Delta H = \int C_p dT$$

这两个公式不受限是否等容、等压

$$U = f(T); \quad H = f(T)$$

$$C_{p,m} - C_{V,m} = R$$

$$C_V = f(T); \quad C_p = f(T)$$





State functions:

Internal energy (U)

$$\Delta U = w + q \quad (1)$$

Enthalpy (H)

$$H = U + PV \quad (2)$$

The equation (2) is always established, no matter whether the pressure is constant or not. But only if the pressure is constant, the following equation (3) can be used.

$$\Delta H = q_p \quad (3)$$

Heat capacity (C)

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V$$

$$C_p = \left(\frac{\partial H}{\partial T} \right)_p$$



Adiabatic process

$q=0$, $\Delta U=w$, p , V , T changing

The change in pressure resulting from an adiabatic, reversible expansion of a perfect gas:

$$pV^\gamma = \text{constant}$$

The heat capacity ratio is defined as:

$$\gamma = \frac{C_{p,m}}{C_{V,m}}$$

Real gases: Joule-Tomson effect

Joule-Tomson coefficient $\mu = \left(\partial T / \partial p\right)_H$

- State function
- 等焓过程，压力-温度的变化情况
- $\mu > 0, p \downarrow T \downarrow$ *positive effect*
- $\mu < 0, p \downarrow T \uparrow$ *negative effect*
- For perfect gas, no effect $\mu = 0$

Isenthalps & inversion curve



Thermochemistry

The standard enthalpy change ($\Delta H^\ominus$), the change in enthalpy for a process in which the initial and final substances are in their standard states.

The standard state: of the pure form a substance at 1 bar and at a specified temperature. Conventionally, $T=298.15\text{ K}$ ($25\text{ }^\circ\text{C}$)



$$\Delta_{trs}H^{\ominus}$$

Physical change

$$\Delta_rH^{\ominus}$$

Chemical change

$$\Delta_fH^{\ominus}$$

The standard enthalpy of formation ($\Delta_fH^{\ominus}$)

The standard reaction enthalpy for the formation of the compound from its elements in their reference states.

The reference state of an element is its most stable state at the specified temperature and 1 bar.



$$\Delta_r H^\ominus = \sum_{\text{Products}} v \Delta H_f^\ominus - \sum_{\text{Reactants}} v \Delta H_f^\ominus$$

$$\Delta_r H^\ominus = \sum_J v_J \Delta H_f^\ominus$$

$$\Delta_r H^\ominus(T_2) = \Delta_r H^\ominus(T_1) + \int_{T_1}^{T_2} \Delta_r C_p^\ominus dT$$

It is normally a good approximation to assume that $\Delta_r C_p$ is independent of the temperature over reasonably limited ranges.

$$\Delta_r H^\ominus(T_2) = \Delta_r H^\ominus(T_1) + (T_2 - T_1) \Delta_r C_p^\ominus$$